POWER PLANT ENGINEERING

Subject Code: 23A03702e

UNIT I: Sources of Energy, Power Plant Economics and Environment

I. Introduction to Sources of Energy

- ***Energy*** = capacity to do work. Power = rate of doing work (kW, MW). Energy = power x time (kWh).
- ***Power plant*** = an assembly of systems / equipment that converts a primary (natural) energy source into ***electrical energy*** on a large scale.
- Primary sources: fuels (coal, oil, gas), water, wind, sun, nuclear fuel, biomass, tides, geothermal.
- Electrical energy is a ***secondary energy*** -- easy to transmit, clean at the point of use, easy to convert to light / heat / motion.

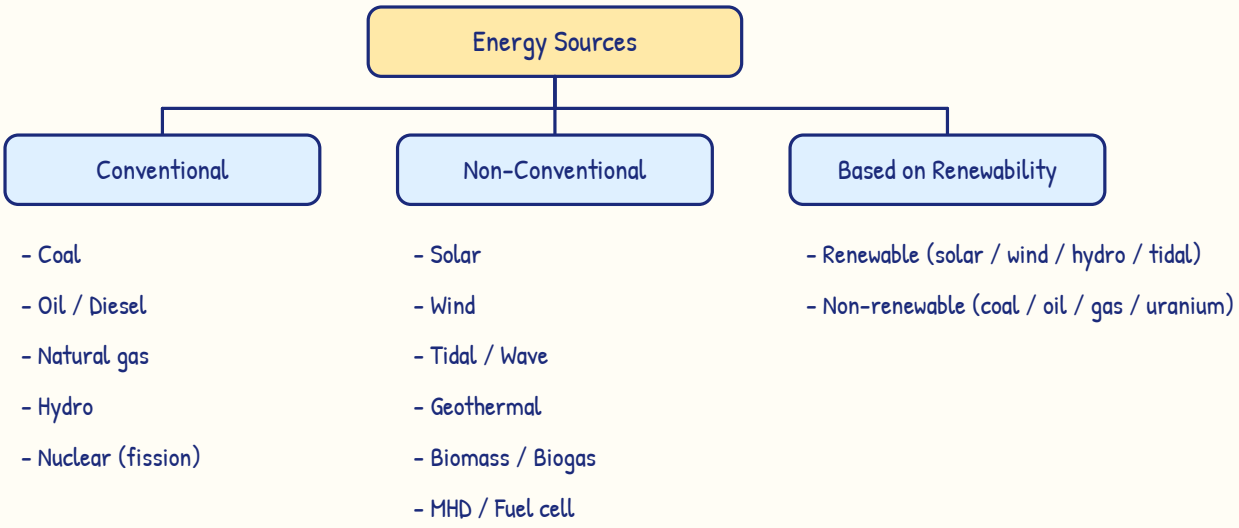


Fig: Classification of energy sources

Conventional vs Non-conventional sources

Point	Conventional	Non-conventional
Nature	Mostly exhaustible (fossil, nuclear)	Mostly renewable, inexhaustible
Examples	Coal, oil, gas, hydro, nuclear	Solar, wind, tidal, biomass, geothermal
Capacity	Large-scale, base-load plants	Small / medium, scattered, intermittent
Pollution	High (CO_2, SO_x, NO_x, ash)	Little or none

Point	Conventional	Non-conventional
Capital cost	Medium to high	Generally high per kW (low load factor)
Running cost	Fuel cost is large	Fuel is free; maintenance only

Why non-conventional sources?

- Fossil fuel reserves are **limited** and costly; import of crude oil burdens the economy.
- Environmental problems: global warming, acid rain, smog.
- Suitable for remote villages where grid extension is uneconomical.

2. Resources and Development of Power in India

- India's installed capacity is about **4.5 lakh MW** (approx.), with **coal / lignite ~ 50%** of it; renewables have the fastest growth.
- **Coal**: reserves mainly in Jharkhand, Odisha, Chhattisgarh, West Bengal, MP -- mostly **non-coking, high-ash (30-45%), low sulphur**, GCV about 3500-5000 kcal/kg.
- **Oil and gas**: Mumbai High, Assam, Krishna-Godavari basin, Gujarat; part imported as LNG.
- **Hydro potential**: about 1,45,000 MW (at 60% load factor); Himalayan rivers, Western Ghats; developed share is only about one-third.
- **Nuclear**: uranium (Jharkhand, Andhra), **thorium** (Kerala monazite sands -- among the largest in the world). Plants: Tarapur, Kalpakkam, Kaiga, Kudankulam, Rawatbhata.
- **Solar**: 300 sunny days a year, 4-7 kWh/m^2/day (Rajasthan, Gujarat). **Wind**: Tamil Nadu, Gujarat, Karnataka, Maharashtra.

Development of power (milestones)

- 1897 Darjeeling hydro station; 1902 Sivasamudram (Karnataka) -- first major hydro plant in India.
- 1948 Electricity (Supply) Act; 1950s: Five Year Plans -- State Electricity Boards (SEBs) formed.
- 1975 NTPC and NHPC; 1969 **Tarapur** nuclear unit; 1989 Power Grid Corporation (PGCIL) -- **National Grid**.

- 2003 Electricity Act (competition, open access); today a ***One Nation, One Grid*** system with 5 regional grids synchronised.
* Power sector is run by Centre (NTPC, NHPC, NPCIL, PGCIL), States (GENCOs, DISCOMs) and the private sector.

3. Power Plant Economics

- A good power plant must supply ***reliable, cheap and sufficient*** power. Economics decides the type of plant, size, and location.

3.1 Capital cost

- Initial investment for land, buildings, boiler / turbine / generator, civil works, transmission, installation and ***interest during construction***.
- Typical cost per kW: Hydro (highest, dams) > Nuclear > Steam > Gas turbine > Diesel (lowest).

3.2 Investment (fixed) charges

- Charges that are ***independent of units generated*** (paid even if the plant is idle):
  - ***Interest*** on capital (the biggest part, say 8–12%)
  - ***Depreciation*** (loss of value with time)
  - Taxes and insurance
  - Salaries of permanent staff, rent

Depreciation methods

- ***Straight line:*** annual depreciation = (P − S) / n, with P = first cost, S = salvage value, n = life in years.
- ***Diminishing balance:*** fixed % of the book value each year.
- ***Sinking fund:*** equal annual deposit q so that it accumulates to (P − S) at interest i: $q = (P - S) \times i / ((1 + i)^n - 1)$

3.3 Operating costs

- ***Fuel cost*** (largest for thermal plants), operating labour, lubricants and water, maintenance and repairs, supplies. They vary with the ***units generated***.
* Total cost per year = Fixed cost (Rs/kW year) + Running cost (Rs/kWh) x Energy generated

4. General Arrangement of Power Distribution

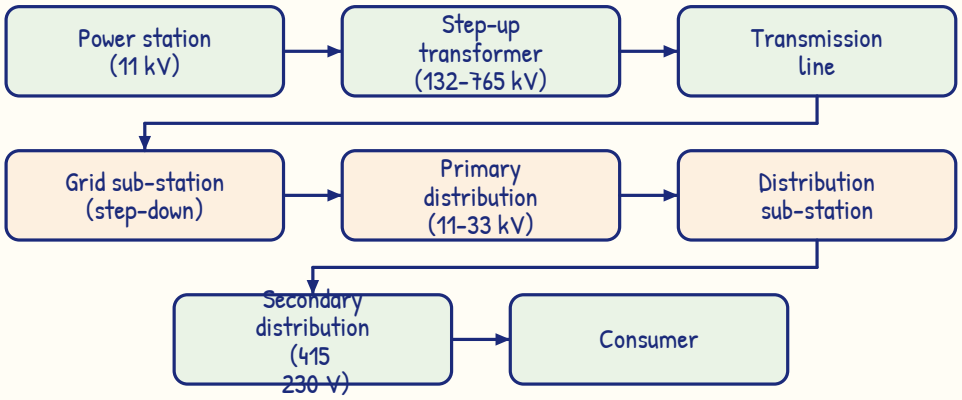


Fig: Path of electric power from station to consumer

- **Generation** at 11 / 13.8 / 21 kV -> stepped up to reduce current and I^2R **loss** during long transmission.
- ***Primary transmission*** 220 / 400 / 765 kV; ***secondary transmission*** 66 / 132 kV; ***primary distribution*** 11 / 33 kV; ***secondary distribution*** 3-phase 415 V, single-phase 230 V.
- Distribution systems: ***radial*** (simple, cheap, poor reliability), ***ring main*** (feeds from two sides, more reliable) and ***interconnected / grid*** (best reliability, common in cities).

5. Load Curves

- ***Load curve*** = graph of load (kW / MW) against time for a given period (day, month or year).
- Shows: peak load, base load, variation, and (from the area) energy generated.

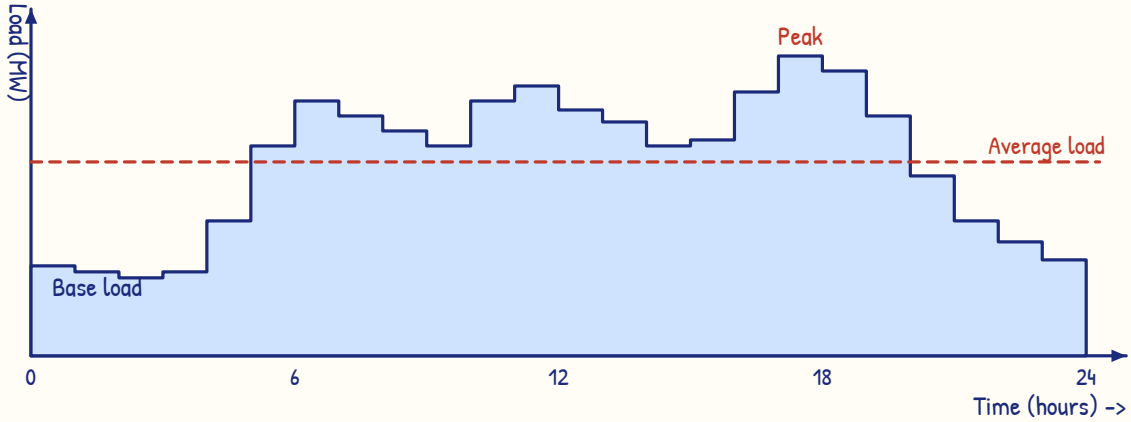


Fig: Daily load curve (chronological)

- Area under daily load curve = ***units generated per day*** (kWh).

- Average load = area / 24 hours. Peak load is usually at evening (lighting + domestic) in India; also a morning peak.

Uses of load curve

- Select the ***number and size of generating units*** and decide on their operating schedule.
- Estimate cost of generation; plan ***maintenance*** during off-peak hours.
- Decide ***base-load and peak-load plants*** (e.g. nuclear / coal = base; hydro / gas turbine / diesel = peak).

Load Duration Curve (LDC)

- Loads are ***rearranged in descending order*** of magnitude, against the number of hours (that load persists). It is always a falling curve.

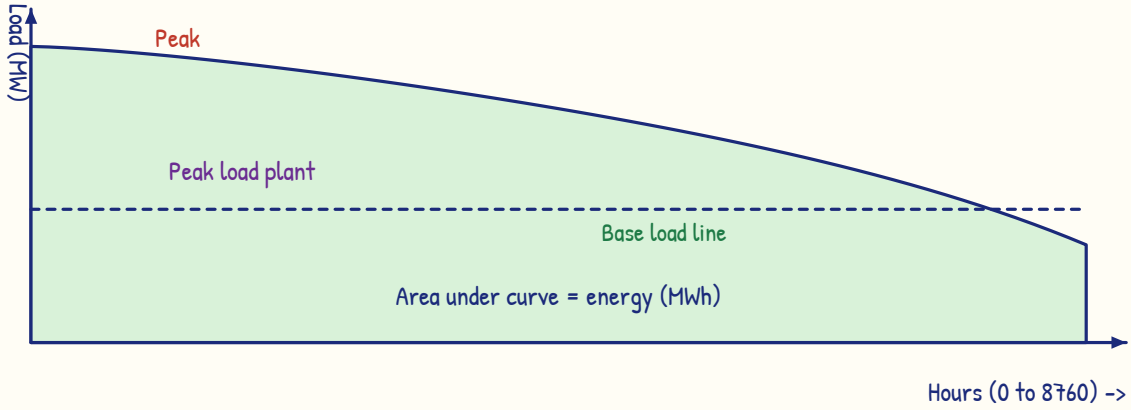


Fig: Load duration curve

- Area under the LDC = ***total energy generated*** (same as the area under the chronological load curve).
- Helps in deciding the ***sharing of load*** between base and peak plants, and in finding capacity factor and load factor easily.

* Chronological load curve shows WHEN the load occurs; the LDC shows HOW LONG each load lasts.

6. Definitions and Factors

Basic definitions

- ***Connected load:*** sum of the continuous ratings of all the equipment connected to the system (kW).
- ***Maximum demand (MD):*** the greatest demand of load on the station during a given period (usually 15 or 30 min average).

- **Average load (demand):** units generated in a period / hours in the period.
- **Installed capacity:** sum of the ratings of all generators in the station.
- **Plant capacity / Reserve:** Reserve capacity = installed capacity - peak load.

Factor	Definition	Formula
Demand factor	Max demand / connected load (always < 1)	MD / Connected load
Load factor	Average load / maximum demand	Units generated / (MD x hours)
Diversity factor	Sum of individual max demands / simultaneous max demand of the group (always >= 1)	Sum of MD_i / MD_{group}
Plant capacity factor	Actual energy produced / (installed capacity x hours)	Avg load / installed capacity
Plant use factor	Actual energy / (capacity x hours the plant was operating)	Energy / (cap x hrs in use)
Utilisation factor	Max demand on the plant / plant capacity	MD / capacity

- **Load factor** is high for a good, economical plant -> cost per unit falls because fixed charges are spread over more units.
- **Diversity factor** high -> less installed capacity needed for the same consumers -> **lower cost** of supply.

* Always remember: Load factor <= 1, Demand factor <= 1, Diversity factor >= 1, Capacity factor <= Load factor (for a plant with reserve).

Relationship

Plant capacity factor = Load factor x (Max demand / Installed capacity) = Load factor x Utilisation factor

- **Reserve** : hot reserve (running, ready), spinning reserve (synchronised, running), cold reserve (stand-by, can be started on demand).

7. Tariff (Cost of Electrical Energy)

- **Tariff** = the rate at which electrical energy is sold to consumers. A good tariff should **recover** fixed + running + profit, be simple, and be fair to different consumers.
- Total cost is divided into: (a) **fixed cost** (depends on max demand / capacity), (b) **semi-fixed** (depends on connected load / customers), (c) **running cost** (depends on kWh).

Types of tariffs

Type	Expression	Remarks
Flat rate	Rs x per kWh	Same for all consumers; simple, unfair
Block rate	First 100 units at Rs a, next at Rs b	Decreasing price with consumption
Two-part	Rs/year = a x kW + b x kWh	a = fixed (kW), b = running; **Hopkinson** tariff
Maximum demand tariff	Charge on MD + energy used	Needs a MD meter; for industries
Three-part	Rs = a + b x kW + c x kWh	a = fixed per customer (billing, meter reading)
Power factor tariff	Rebate / penalty with p.f.	Encourages high p.f. (>= 0.9)
Time-of-day (ToD)	Higher rate at peak hours	Reduces peak, flattens load curve
Wright / Doherty	Wright: kW x connected-load basis; Doherty: tariff by use	Rarely used now

- Low power factor -> more current for the same power -> bigger cables / transformers, more loss; so utilities charge a **penalty**.

8. Related Exercises (worked)

Example 1: Load factor and capacity factor

- A plant of **installed capacity 100 MW** produces **4.9 x 10^8 kWh** in a year. Maximum demand = **80 MW**.
- Average load = 4.9 x 10^8 / 8760 = **55.9 MW**
- Load factor = 55.9 / 80 = **0.70 (70%)**; Capacity factor = 55.9 / 100 = **0.559 (55.9%)**

Example 2: Diversity factor

- A station supplies loads with maximum demands 8 MW, 10 MW, 12 MW and 15 MW. The simultaneous maximum demand of the station is 30 MW.
- Diversity factor = (8 + 10 + 12 + 15) / 30 = 45 / 30 = **1.5**

Example 3: Daily units and load factor

- A power station has a **daily load**: 0–6 h : 20 MW, 6–12 h : 60 MW, 12–18 h : 40 MW, 18–24 h : 80 MW.
- Energy / day = 6(20 + 60 + 40 + 80) = 6 x 200 = **1200 MWh**; Average load = 1200 / 24 = **50 MW**
- Max demand = 80 MW -> Load factor = 50 / 80 = **0.625 = 62.5%**

Example 4: Two-part tariff

- A consumer has a MD of 100 kW and a load factor of 40%. Tariff: Rs 200 per kW of MD per year + Rs 3 per kWh.
- Energy per year = 100 x 0.4 x 8760 = **3,50,400 kWh**
- Annual bill = 200 x 100 + 3 x 3,50,400 = 20,000 + 10,51,200 = **Rs 10,71,200**; Average rate = Rs 3.06 per kWh

Example 5: Depreciation (straight line)

- A boiler costs Rs 50 lakh, salvage value Rs 5 lakh, life 15 years. Annual depreciation = (50 − 5)/15 = **Rs 3 lakh per year**.

Base-load and peak-load stations

- **Base-load plants** run continuously at high load factor: nuclear, large coal / lignite, run-of-river hydro. Low running cost, slow to start / change load.
- **Peak-load plants** run only a few hours a day: diesel, gas turbine, pumped storage, storage hydro. Quick start (minutes), high running cost.

- **Intermediate / cycling plants:** medium coal and combined-cycle units following the daily variation.
- **Interconnected (grid) operation** gives: better load factor and diversity, less reserve capacity, higher reliability and lower cost per unit; the most efficient plants are loaded first (**merit order dispatch**).

Example 6: Choosing plant capacity

- Annual peak demand of a city is 120 MW, load factor 55%, the reserve required is 15% of peak. Find units supplied per year and installed capacity.
- Energy = 120 x 0.55 x 8760 = **5,78,160 MWh (5.78×10^8 kWh)**
- Installed capacity = 120 x 1.15 = **138 MW**; Capacity factor = 66 / 138 = **47.8%**

Example 7: Annual cost per kWh

- Plant 50 MW, capital cost Rs 1000 crore @ interest + depreciation 12% p.a.; load factor 60%, running cost Rs 2.5 per kWh.
- Energy = 50 x 1000 x 0.6 x 8760 = **26.28×10^7 kWh**; Fixed charges = 0.12 x 1000 crore = Rs 120 crore
- Fixed cost per kWh = 120×10^7 / 26.28×10^7 = **Rs 4.57**; Total = 4.57 + 2.5 = **Rs 7.07 / kWh**
- If load factor rises to 80%, fixed cost falls to Rs 3.42 -- shows **why a high load factor reduces cost**.

9. Effluents from Power Plants and Impact on Environment

- Every power plant gives out **effluents** (solid, liquid, gaseous, thermal, radioactive) -- need control to protect health and nature.

Plant	Main effluents	Impact
Thermal (coal)	Fly ash, bottom ash, SO_2, NO_x, CO_2, CO, particulates (SPM), hot water	Acid rain, smog, global warming, respiratory disease, thermal pollution of rivers
Diesel / gas	NO_x, CO, unburnt HC, soot, noise	Smog, ozone formation, hearing loss

Plant	Main effluents	Impact
Hydro	No emissions; reservoir submergence	Loss of land and forest, displacement, silting, flora / fauna change, methane from rotting plants
Nuclear	Radioactive waste, hot water	Radiation hazards (cancer, mutation); long-lived waste disposal

Environmental effects in short

- **Greenhouse effect / global warming:** CO_2, CH_4, N_2O trap heat -> sea level rise, climate change.
- **Acid rain:** SO_2 + NO_x + water -> H_2SO_4 and HNO_3; damages crops, buildings, lakes.
- **Particulate matter:** PM_{10} and $PM_{2.5}$ penetrate lungs; fly ash dumping occupies land.
- **Thermal pollution:** warm condenser water discharged raises river temperature -> dissolved oxygen drops -> aquatic life dies.
- **Ozone layer depletion** and **photochemical smog** due to NO_x and HC.

10. Pollutants and Pollution Standards

Types of pollutants

- **Primary pollutants:** emitted directly -- CO, SO_2, NO, particulates, hydrocarbons.
- **Secondary pollutants:** formed in the atmosphere -- ozone, PAN, acid mist, sulphuric acid.

- In India, standards are set by the **CPCB** (Central Pollution Control Board) / State PCBs under the Air (Prevention and Control of Pollution) Act 1981, Water Act 1974 and Environment (Protection) Act 1986.

Parameter	Typical standard / limit
Particulate matter (new coal plants)	30 mg / Nm^3 (revised) ; older units 50–150 mg/Nm^3

Parameter	Typical standard / limit
SO_2	200 mg/Nm^3 (units up to FGD norms) ; stack height 220 m for 500 MW and above
NO_x	300 mg/Nm^3
Mercury	0.03 mg/Nm^3
Condenser cooling water discharge	Temperature rise not more than 5 deg C above intake
Ambient air (NAAQS) SO_2	80 micro-g/m^3 (24 hr) ; PM_{10} : 100 ; $PM_{2.5}$: 60

* Limits are revised from time to time (2015, 2018 notifications) -- remember the idea: ***tall stacks + particulate removal + FGD + low-NO_x burners***.

II. Methods of Pollution Control

Control of particulates

- ***Mechanical collectors / cyclone separators*** -- for coarse dust.
- ***Electrostatic precipitators (ESP)*** -- charge dust particles, collect on plates; efficiency ***99%+***.
- ***Bag (fabric) filters*** and ***wet scrubbers*** -- fine particles.

Control of gaseous emissions

- ***SO_2***: low-sulphur coal, coal washing, ***flue gas desulphurisation (FGD)*** with limestone slurry ($CaCO_3$ + SO_2 -> $CaSO_3$ -> gypsum), fluidised bed combustion with limestone.
- ***NO_x***: low-NO_x burners, staged / lower-temperature combustion, flue-gas recirculation, ***selective catalytic reduction (SCR)*** with ammonia.
- ***CO_2***: higher plant efficiency (supercritical, IGCC), carbon capture, afforestation, renewables.
- ***Tall stacks*** disperse gases over a wide area (dilution, not removal).

Control of water and thermal pollution

- ***Cooling towers*** / cooling ponds instead of once-through cooling; treat effluent and ash-pond overflow (neutralise, settle).

- **Ash:** use fly ash in cement, bricks, road embankments -- also dry ash handling.

Radioactive waste

- Store the spent fuel in water pools -> reprocess -> vitrify and bury in deep geological repositories. Shielding and monitoring with dosimeters.

Noise

- Silencers, acoustic enclosures for diesel sets, ear-muffs for workers.

12. Inspection and Safety Regulations

- Governed by the **Indian Electricity Act 2003**, **Indian Boiler Act 1923** / Indian Boiler Regulations (IBR), the **Factories Act 1948**, and CEA safety regulations.
- **Inspection:** boilers, pressure vessels and pipes are inspected by the **Chief Inspector of Boilers** -- hydraulic test at 1.5 x working pressure, periodic (yearly) inspection and certificate before use.
- Safety valves, water-level gauges, pressure gauges, fusible plugs, feed check valves and blow-down valves must be fitted and tested.

Safety rules in a power plant

- Earthing of all equipment; **lock-out / tag-out** (permit to work) before maintenance.
- Fire protection: fire hydrants, CO_2 / foam extinguishers, **fire walls** and deluge systems for transformers, oil storage and coal bunkers (spontaneous combustion).
- Personal protective equipment: helmets, gloves, goggles, ear plugs, safety shoes.
- Proper ventilation, safe handling of coal dust (explosion risk) and hazardous chemicals (hydrogen, chlorine).
- Emergency shut-down procedures, trained staff, first aid and regular **mock drills**; log books and records kept up to date.
- Nuclear plants: **AERB** (Atomic Energy Regulatory Board) licensing, radiation zones, dose limits and interlocks.

Quick revision

- Load factor = Avg load / Max demand. Demand factor = MD / Connected load. Diversity factor = Sum of MD / Group MD.
- Fixed charges: interest, depreciation, taxes. Running charges: fuel, labour, maintenance.
- Two-part tariff = Rs/kW + Rs/kWh. ESP removes dust; FGD removes SO_2; SCR removes NO_x.

POWER PLANT ENGINEERING

Subject Code: 23A03702e

UNIT II: Steam Power Plant: Boilers, Fuel Handling and Combustion

1. Steam Power Plant – Introduction

- A ***steam power plant (thermal station)*** converts the chemical energy of fuel (coal) into heat -> steam -> mechanical work -> electricity, using the ***Rankine cycle***.
- Four main circuits: (1) ***coal and ash circuit***, (2) ***air and flue gas circuit***, (3) ***feed water and steam circuit***, (4) ***cooling water circuit***.

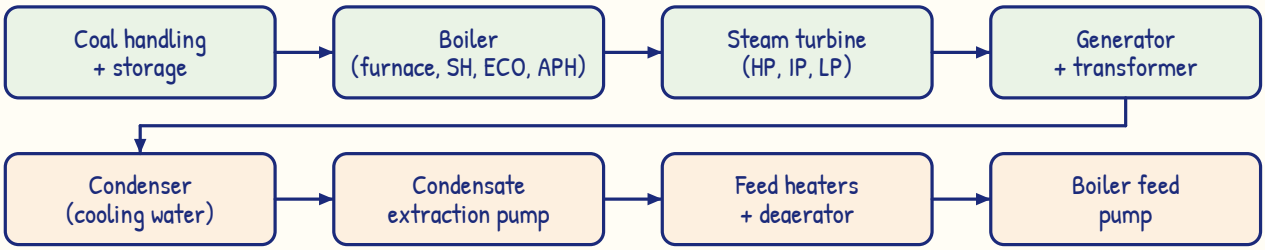


Fig: Layout of a steam power plant

- Flue gas path: furnace -> superheater -> reheater -> ***economiser*** -> ***air pre-heater*** -> dust collector (ESP) -> ***ID fan*** -> chimney.
- Advantages: large power, fuel is cheap, reliable, less initial cost than hydro. Disadvantages: pollution, high running cost, long start-up, needs lot of water.

2. Boilers – Introduction

- ***Boiler (steam generator)*** = closed vessel in which water is converted to steam at constant pressure by heat of combustion.

Classification

Basis	Types
Tube contents	***Fire-tube*** (hot gas inside tubes: Lancashire, Cochran, locomotive) and ***Water-tube*** (water inside tubes: Babcock-Wilcox, Stirling)
Circulation	Natural, forced (La Mont), once-through (Benson, Sulzer)
Pressure	Low (< 20 bar), medium (20–80), high (> 80 bar), supercritical (> 221 bar)

Basis	Types
Use	Stationary (power), portable, marine, locomotive

- Power stations use **water-tube boilers** -- large capacity, high pressure, quick steaming, but need good feed-water quality.

Fire-tube vs water-tube

- Fire-tube: low pressure (<= 25 bar), low output, simple, cheap, large water content, less sensitive to water quality.
- Water-tube: high pressure (up to 300+ bar), large output, high efficiency, small water content, lighter, quick start, needs pure feed water.

Boiler mountings and accessories

- **Mountings** (safety): safety valve, water-level gauge, pressure gauge, stop valve, feed check valve, blow-off cock, fusible plug, manhole.
- **Accessories** (efficiency): **economiser** (heats feed water with flue gas), **air pre-heater**, **superheater**, **reheater**, feed pump, injector, soot blower.

3. Modern High-Pressure and Supercritical Boilers

- Raising pressure and temperature of steam raises Rankine efficiency. Modern drums use natural / controlled circulation up to ~ 180 bar.
- **High-pressure boilers** (above 80 bar) use forced circulation and small-bore tubes; examples: **La Mont**, **Velox**, **Benson**, **Loeffler**.

La Mont boiler (forced circulation)

- A **centrifugal pump** circulates water from the drum through the evaporator tubes at about **8-10 times** the steam generated -> good heat transfer, no steam blanketing. Working pressure ~ 120-150 bar.

Benson boiler (once-through)

- No drum. Feed water passes in one pass through economiser -> evaporator -> superheater; works at **critical / supercritical pressure** (> 225 bar), so water changes to steam without bubbling (no latent heat phase).
- Efficient, light weight, quick start; needs very pure water, no blow-down, complex control.

Loeffler and Velox

- **Loeffler:** steam, not water, circulated (by pump) through the evaporator drum; avoids scale. **Velox:** supercharged combustion (gas turbine compressor drives gas at ~ 2.5 bar); compact, high heat release.

Supercritical technology

- **Critical point of water: 221.2 bar, 374.15 deg C.** Above it, no distinction between liquid and vapour -> no boiling, no drum.
- **Supercritical:** 240-250 bar, 540-565 deg C; **Ultra-supercritical (USC):** 270-300 bar, 600-620 deg C; **Advanced USC:** 700+ deg C (nickel alloys).
- Plant efficiency improves from ~ 35% (subcritical) to **40-42% (SC)** and **44-46% (USC)** -> less coal, less CO_2.
- Indian units: 660 MW and 800 MW supercritical sets (Sipat, Mundra, Barh, Sagardighi).
- * Challenges: high-temperature metallurgy, feed water purity, accurate control, thick walls (thermal stress).

4. Analysis of Power Plant Cycles

- Ideal cycle for steam plants = **Rankine cycle**: boiler -> turbine -> condenser -> pump.

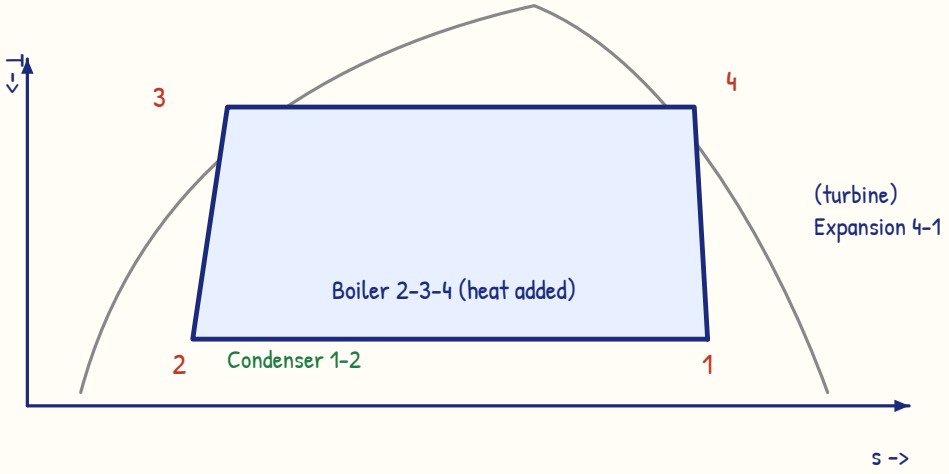


Fig: Rankine cycle on the T-s diagram (1: turbine exit, 2: pump exit, 3-4: boiler)

Processes

- 1-2: condensate pumping (adiabatic, reversible, small work). 2-3-4: heat addition in boiler (water heating, boiling, superheating at constant pressure).
- 4-1: **isentropic expansion** in turbine; 1 -> 2 : constant-pressure heat rejection in condenser (typically 0.04-0.1 bar abs).

Rankine cycle formulae
• Turbine work $W_T = h_4 - h_1$; Pump work $W_P = h_2 - h_1' = v \times (p_b - p_c)$

- Heat supplied $Q_1 = h_4 - h_2$; **Efficiency** = $(W_T - W_P) / Q_1$
- **Specific steam consumption** (kg/kWh) = 3600 / $(W_T - W_P)$; **Heat rate** = SSC x Q_1 (kJ/kWh)
- Back-work ratio = W_P / W_T (tiny, about 1%)

- Rankine efficiency increases with: **higher boiler pressure, higher superheat temperature, lower condenser pressure**.
- Drawbacks of raising pressure alone: wetter exhaust (low dryness fraction < 0.85) -> erosion of last-stage blades -> solved by **reheat**.

Reheat cycle

- Steam is taken out after HP turbine, **reheated** in the boiler to (say) 540 deg C, and expanded in IP / LP turbines. Raises dryness at exhaust and cycle efficiency by 4-5%.

Regenerative cycle

- Some steam is **bled** from the turbine to heat the feed water in **feed-water heaters** (open / closed type) -> the average temperature of heat addition rises -> efficiency rises; also helps deaeration.
- With ideal infinite stages, the efficiency approaches **Carnot**. Practical stations use **6-8 heaters**.

Reheat-regenerative cycle (modern practice)

- Combines both: superheat, **single reheat** (double in large units), 7-8 bleed points. Overall thermal efficiency 38-45%.

Binary vapour cycle and cogeneration

- **Binary cycle:** mercury (high temp.) + steam (low temp.) -- historic, mercury toxic. **Cogeneration (CHP):** back-pressure / extraction turbines supply process steam + power; fuel utilisation 70-85%.

5. Modern Trends in Cycle Improvement and Waste Heat Recovery

- Increase **steam parameters** (SC / USC), reheat, more regenerative heaters, low condenser pressure with good cooling.
- **Combined cycle (gas + steam)** efficiency 55-60%; **IGCC** (integrated gasification combined cycle) for coal.
- **Variable-pressure operation**, better turbine blade design, **digital controls** and condition monitoring, drives with **variable-frequency** drives for pumps and

fans.

Waste heat recovery (WHR)

- About 10–15% of energy is lost in flue gas and the rest in the condenser. Recovered by:
  - **Economiser** -- feed water heated with flue gas (every 20 deg C drop in gas temperature gives ~ 1% fuel saving).
  - **Air pre-heater** -- combustion air heated by flue gas (recuperative tubular type, **regenerative Ljungstrom** rotary type).
  - **Waste heat recovery boiler (HRSG)** in combined cycle and process industries.
  - Condenser heat for district heating / desalination; blow-down heat recovery; **steam jacketing** and insulation.
- Efficiency improvement of a boiler: **dry flue gas loss**, moisture loss and unburnt carbon loss are the main losses; boiler efficiency 85–90% (on GCV).

6. Fluidised Bed Boilers

- Coal (crushed, 6–10 mm) + limestone are supported on a **bed of ash / sand**; air from below at velocity 1–3 m/s makes the bed behave like a **boiling liquid** (fluidised) -> intense mixing.



Fig: Principle of a fluidised-bed boiler

Features

- Bed temperature **800–950 deg C** (below ash fusion point) -> **low NO_x** (thermal NO_x negligible).
- **SO_2 captured in the bed** by limestone ($CaCO_3$ -> CaO; $CaO + SO_2 + 1/2\ O_2$ -> $CaSO_4$) -- 90% removal with no FGD.
- Burns **low-grade, high-ash and multiple fuels** (coal washery rejects, biomass, petcoke, lignite). High heat-transfer, compact.

Type	Description
Bubbling FBC (BFBC)	Low velocity (1–3 m/s); distinct bed surface; in-bed tubes; for small units
Circulating FBC (CFBC)	High velocity (4–8 m/s); solids carried out, caught in cyclone and returned; suited to 125–300 MW

Type	Description
Pressurised FBC (PFBC)	Bed at 10–16 bar; hot gases drive a gas turbine -> combined cycle

- Drawbacks: higher auxiliary power (fans), erosion of tubes, limestone handling, N_2O emission.

7. Fuel and Handling Equipment – Types of Coals

- **Coal** forms from plants by coalification: Peat -> Lignite -> Sub-bituminous -> Bituminous -> Anthracite (carbon content and calorific value increase).

Type	Fixed carbon	Volatile matter	GCV (kcal/kg)	Remarks
Peat	40–55%	high	3500–4500	Early stage, 80% moisture
Lignite	60–70%	30–40%	4000–5500	High moisture, brown, burns easily
Bituminous	70–86%	15–40%	6500–8000	Best for power / coking, soft black
Anthracite	86–98%	3–8%	7800–8200	Hard, smokeless, hard to ignite

- Indian coals: mostly **non-coking, ash 30–45%**, moisture 4–10%, GCV about 3500–5500 kcal/kg.
- **Proximate analysis:** moisture, volatile matter, fixed carbon, ash. **Ultimate analysis:** C, H, O, N, S (% by weight). Higher calorific value (GCV) = LCV + latent heat of moisture.
- Dulong: **GCV = 33,800 C + 1,44,000 (H – O/8) + 9,270 S kJ/kg**.
- Fuel oil and natural gas are also used for start-up and flame support.

8. Coal Handling

- Large stations burn **10,000–20,000 tonnes of coal per day**; the handling system must be **reliable, continuous and dust-free**.

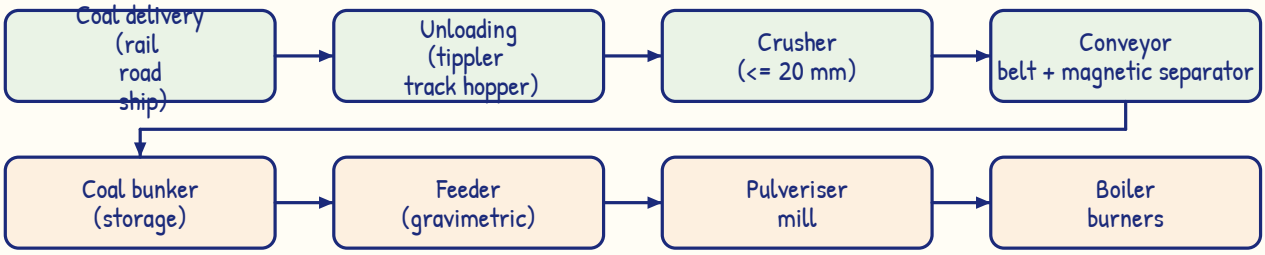


Fig: Coal handling sequence

Stages

- **Delivery:** railway wagons (BOXN), trucks, ships, MGR (merry-go-round) system.
- **Unloading:** wagon tippler, bottom-discharge hopper, car shaker, grab bucket crane.
- **Preparation:** crushers (jaw, hammer, ring-granulator), screens, **magnetic separators** (remove iron), sampling and weighing (**belt weigher**).
- **Transfer:** belt conveyor, screw conveyor, bucket elevator, skip hoist, chain / apron conveyor.
- **Storage:** bunkers (daily) and open stock-yard (live and dead storage).

Choice of handling equipment

Equipment	Merits	Limits
Belt conveyor	Large capacity (up to 5000 t/h), long distance, little power, quiet, cheap	Inclination up to ~ 18–20 deg only; dust
Screw conveyor	Simple, enclosed, low cost	Wear, short distance, small quantity
Bucket elevator	Vertical lift, compact	Limited capacity, wear
Skip hoist	Vertical / steep lift	Intermittent, higher cost
Apron / chain	Handles hot / lumpy coal, heavy duty	Costly maintenance

* Choice depends on: capacity required, distance and elevation, nature (size, moisture) of coal, space, capital + running cost and reliability.

Coal storage

- **Live storage** (for 1 week, near the boiler) and **dead storage** (30 – 60 days stock, compacted, kept as a reserve).
- Problems: **spontaneous combustion** (oxidation heating), loss of calorific value, wind loss, fire. Prevent by compacting in layers, side slopes, ventilation + temperature probes, keeping separate coal types, and **first in first out** use.

- Under-water storage avoids oxidation (cost + wetting issues).

9. Ash Handling Systems

- Coal with 40% ash gives huge ash: a 500 MW station may give **> 6000 t/day**. Ash = ***bottom ash*** (clinker, 20%) + ***fly ash*** (80%, in pulverised firing).
- Ash is ***abrasive, hot and dusty*** -- the handling system should be safe, efficient and economical.

System	Principle	Remarks
Mechanical	Belt, bucket or scraper conveyor	Small plants; high wear
Hydraulic	Ash sluiced with water to a sump / ash pond (slurry pump)	Common, large plants; large water demand, pollution
Pneumatic	Ash conveyed in pipes by vacuum (negative) or pressure (positive) air	Dry fly ash; clean; for ESP hoppers; long distance
Steam jet	Steam jet carries ash in a pipe	Small; high steam use

- ***Dense-phase*** pneumatic conveying and ***high-concentration slurry disposal (HCSD)*** are used in modern plants to save water.
- Utilisation: cement, bricks, concrete blocks, road embankments, land-fill, mine filling (Indian policy: ***100% utilisation***).

10. Combustion Process – Properties of Coal

- **Combustion** = rapid chemical reaction of fuel with oxygen, releasing heat. **Three T's: Time, Temperature, Turbulence.**
- Coal burns in stages: **drying -> devolatilisation (volatile matter burns) -> char (fixed carbon) burning -> ash.**

Basic combustion reactions

- $C + O_2 \rightarrow CO_2 + 33{,}800$ kJ/kg; $2C + O_2 \rightarrow 2CO$ (incomplete)
- $2H_2 + O_2 \rightarrow 2H_2O + 1{,}44{,}000$ kJ/kg; $S + O_2 \rightarrow SO_2 + 9{,}270$ kJ/kg
- **Theoretical air** for 1 kg coal = 11.6 C + 34.8 (H – O/8) + 4.35 S (kg of air). **Excess air** (20–30% for coal) is given for complete combustion.

Important properties of coal for burning

- **Calorific value**, volatile matter (easy ignition, long flame), fixed carbon, moisture (lowers CV, ignition difficult), **ash content and ash fusion temperature**, caking and swelling index, grindability (**Hardgrove index**), sulphur content, size.

11. Fuel Beds – Overfeed and Underfeed

- **Overfeed:** fresh coal is fed **on top** of burning bed, air from below; coal passes through zones -> ignition from the hot bed below. (travelling grate, spreader).
- **Underfeed:** fresh coal is fed **from below** the fire by a screw / ram; volatiles rise through the hot fuel bed above and burn completely -> **smokeless**. (retort stoker)

Stokers (mechanical firing)

- Used for boilers up to ~ 100–200 t/h of steam.

Stoker	Description	Features
Travelling grate	Endless chain grate moves from front to rear; coal on grate, air from below in zones; ash dropped at end	20–50 t/h; suits non-caking coal, low ash fusion; ignition arch needed
Spreader	Coal thrown by rotor / flipper onto grate; fine burns in suspension, coarse on grate	Quick response, handles many coals; fly-ash high, needs dust collector

Stoker	Description	Features
Retort (underfeed)	Coal pushed into retort by screw, rises, air through tuyeres; ash overflows on sides	Small boilers, bituminous caking coal; smokeless

- ***Vibrating grate*** and ***chain grate*** are variants. Stokers are limited to modest capacity -> for large units ***pulverised fuel firing***.

12. Pulverised Fuel Burning System

- Coal is ground to powder (***75% through 200 mesh = 75 micron***) and blown by ***primary air*** into the furnace where it burns in suspension like gas -> very high intensity.

Advantages

- Large boilers (> 100 t/h) possible, burns any coal (also low grade), quick response, low excess air (15–20%), high efficiency, less ash on grate.

Disadvantages

- High capital + power for mills, fly-ash and dust pollution (needs ESP), explosion risk, flame stability needs support oil, furnace slagging.

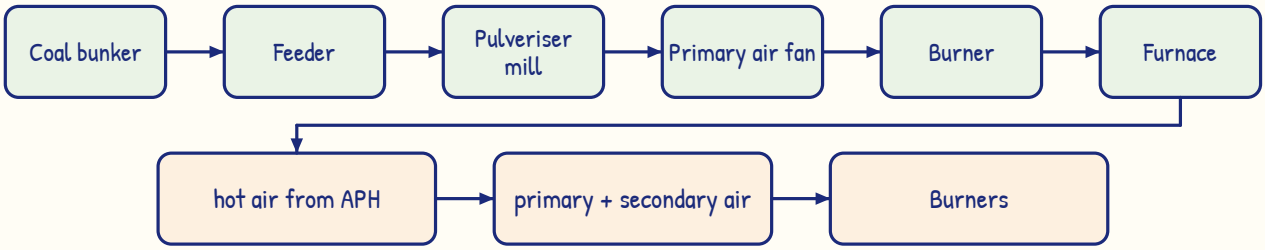


Fig: Pulverised-fuel firing (unit system)

Components

- ***Feeder*** (regulates coal flow), ***pulveriser / mill*** (ball and tube mill -- slow; ***bowl / ring-roll mill*** -- medium; ***hammer / impact mill*** -- high speed, for lignite), ***classifier***, ***primary air fan*** (carries + dries coal), ***burners*** (tangential, wall, or downshot), ***igniters***.
- ***Unit system*** (each mill feeds one boiler directly, common now) vs ***central (bin) system*** (powder stored in bunker, flexible).
- ***Types of firing:*** tangential (corner) firing -- fireball in the furnace; ***front / opposed wall firing*** with swirl burners.
- ***Slag*** removal: ***wet bottom*** (molten ash) and ***dry bottom*** furnace (ash as solid, common).

13. Cyclone Furnace

- A horizontal **water-cooled cylinder** attached to the boiler furnace; crushed coal (**4-6 mm**, not powdered) is fed into the cyclone with **primary air tangentially** (swirl).
- Secondary air enters tangentially at a high speed (~ 100 m/s) -> coal particles are flung to the wall and burn in a **molten slag layer**; heat release ~ 5 x 10^6 kcal / m^3h at **1650-1900 deg C**.
- **Advantages:** crushed coal, less fly ash (**~ 20-30%**; the slag is removed as liquid), small furnace volume, low excess air (10-15%), uses low-fusion-temp ash coals.
- **Disadvantages:** high fan power (high-pressure air), high NO_x, needs coals with low ash fusion temperature, high maintenance.

14. Combustion Needs and Draught System

- Combustion needs a **continuous supply of air** and removal of flue gas -> created by **draught** = pressure difference that causes flow of air / gas.

Types of draught

Type	Created by	Remarks
Natural	Chimney (density difference hot gas vs ambient)	Simple, no power; small; height 20-100 m; not controllable
Induced (ID fan)	Fan after boiler (pressure below atm. in furnace)	Handles hot gas; with chimney; leaks air inwards
Forced (FD fan)	Fan before boiler (furnace above atm.)	Cold air, small fan; hot gas may leak out
Balanced	Both FD + ID fan; furnace slightly negative (-5 to -10 mm WG)	**Used in large plants**; best control

Chimney draught

- Draught head: $h_w = H (\rho_a - \rho_g)$ in metres of gas column; in mm water: $h_w = 353\ H\ (1/T_a - 1/T_g)$ with temperatures in K.
- Maximum draught is obtained when $T_g = 2\ T_a$ (about); practical chimneys have flue gas at 150-200 deg C.

- Height of chimney is also fixed by the **pollution dispersion**: 220 m for > 500 MW units (see Unit I).

- **Fans:** centrifugal (backward curved / radial), **axial fans** for large flows (**with variable pitch**). Flow control: dampers, inlet guide vanes, speed (VFD).

15. Dust Collectors

- Remove **fly ash** from flue gas before the chimney.

Type	Working	Efficiency
Gravity settling chamber	Velocity reduced, dust settles	40–60% (coarse)
Cyclone separator	Centrifugal force in vortex	70–90%
Wet scrubber	Water spray captures particles	90–95%
Fabric (bag) filter	Dust retained on cloth bags	99%+ (fine)
Electrostatic precipitator (ESP)	Dust charged by corona (30–60 kV DC), pulled to collecting plates, rapped off to hoppers	**99–99.9%**

- ESP performance depends on the dust resistivity (10^{7}–10^{10} ohm-cm ideal), gas velocity (1 m/s), and plate area (**Deutsch** equation: eta = $1 - e^{-wA/Q}$).

16. Cooling Towers and Heat Rejection

- Condenser carries heat from exhaust steam (about 55–60% of fuel energy). The warm cooling water is cooled in a **cooling tower** and reused (closed cycle) if water is scarce.
- **Principle:** warm water falls through **fill**; part evaporates and removes latent heat -> cools the rest (can reach near the **wet-bulb** temperature).

Type	Features
Natural draught (hyperbolic)	Large shell 100–200 m, no fan, low running cost, high capital; large power stations
Mechanical draught (induced / forced fan)	Smaller, compact, fan power, controllable; cells in a row
Wet / Dry / Hybrid	Wet: evaporation; Dry: radiator (no loss of water, expensive); Hybrid: both, reduces plume

- Terms: **Range** = hot water temp. – cold water temp.; **Approach** = cold water temp. – wet bulb temp.; **Efficiency** = Range / (Range + Approach).
- Losses: evaporation (~ 1%), drift, blow-down. Make-up water ~ 2–3% of circulating water. Alternatives: cooling ponds, spray ponds, once-through (sea water).

17. Analysis of Pollution from Thermal Power Plants, Controls and CO_2 Recorder

- Pollution: **fly ash (SPM)**, SO_2, NO_x, CO, CO_2, mercury, hot water, noise and coal dust. A 500 MW station may emit ~ **10–15 Mt CO_2 per year** and burns over 7500 tonnes/day of coal.

Pollutant	Source	Control
Particulates	Fly ash	ESP, bag filter, cyclone
SO_2	Sulphur in coal	FGD, low-S coal, FBC with limestone
NO_x	High-temp. combustion	Low-NO_x burners, OFA, SCR / SNCR
CO_2	Combustion	High efficiency, CCS, co-firing biomass
Thermal	Condenser	Cooling tower

- Also: tall chimneys, coal washing, green belt, ash pond lining, ambient air monitoring (see Unit I).

CO_2 recorder

- Used for **flue gas analysis** to check combustion efficiency: CO_2 % is high when excess air is low; the flue-gas CO_2 percentage and the O_2 % tell the **air–fuel ratio**.
- **Orsat apparatus** (chemical absorption): measure a flue-gas sample (100 cc) over water, pass through absorbents: **KOH** absorbs CO_2, **alkaline pyrogallol** absorbs O_2, **cuprous chloride** absorbs CO. The decrease in volume = % of each gas.
- **CO_2 recorder (automatic):** a gas sample is continuously drawn; CO_2 is absorbed in KOH solution -> volume decrease measured and recorded on a chart; based on this the operator adjusts air supply. Modern units use **IR / zirconia O_2 analysers**.
- Typical values: CO_2 = 12–15% in coal flue gas; **Excess air % = $21/(21 - O_2) - 1$** (in %).

Quick revision

- Rankine efficiency up by higher pressure, superheat, reheat, regeneration, lower back-pressure.
- FBC: 800–950 deg C, low NO_x, in-bed SO_2 capture. PF firing: 75 micron, > 100 t/h. Cyclone: crushed coal, slag.
- Balanced draught = FD + ID. ESP 99%+. Cooling tower approach = cold water – WBT.

POWER PLANT ENGINEERING

Subject Code: 23A03702e

UNIT III: Diesel Power Plant and Gas Turbine Plant

1. Diesel Power Plant – Introduction

- A ***diesel power plant*** uses a ***diesel engine (compression-ignition, CI)*** as prime mover to drive an alternator. Fuel is burnt inside the cylinder (internal combustion).
- Rating: from a few kW to about ***50 MW***; typical stations 2–50 MW. Used as ***stand-by***, ***peak-load***, ***emergency*** and ***isolated / remote area*** supply (hospitals, telecom, islands, mines, industries).

Advantages

- Simple design and layout, ***quick start*** (in ~ 1–2 minutes) and stop, small space, low initial cost, ***no stand-by losses***, thermal efficiency 35–40% (higher than a small steam plant), little cooling water, easy to operate, good part-load efficiency.

Disadvantages

- ***High running cost*** (diesel is costly), high maintenance and lubricating oil cost, limited capacity, ***noise*** and vibration, smoky exhaust (NO_x, soot), cannot take overloads for long, fuel storage hazards.

2. Working of Diesel Engine (revision)

- Air alone is ***compressed*** (compression ratio ***14:1 – 22:1***) -> temperature rises to ~ 500–700 deg C; fuel is injected as a fine spray at the end of compression and ***self-ignites*** (no spark plug).

Point	4-stroke engine	2-stroke engine
Cycle completed in	4 strokes (2 revolutions)	2 strokes (1 revolution)
Strokes	Suction, compression, power, exhaust	Compression + (suction / scavenging) combined with power + exhaust
Valves	Inlet + exhaust valves	Ports (scavenge + exhaust)

Point	4-stroke engine	2-stroke engine
Power	Less for same size	Nearly double for same size
Efficiency	Better fuel economy	Some fuel lost in scavenging
Use	Medium / small sets	Large, slow marine / power engines

- Cycle: ideal **Diesel cycle** (heat added at constant pressure); real engines follow the **dual cycle**.

> Diesel cycle efficiency = 1 - (1 / $r^{gamma-1}$) x [(rho^{gamma} - 1) / (gamma (rho - 1))] (r = compression ratio, rho = cut-off ratio)

- **Brake thermal efficiency** = brake power / (fuel flow x calorific value). **Specific fuel consumption** (sfc) = about **0.20–0.25 kg/kWh**.

3. Plant Layout with Auxiliaries

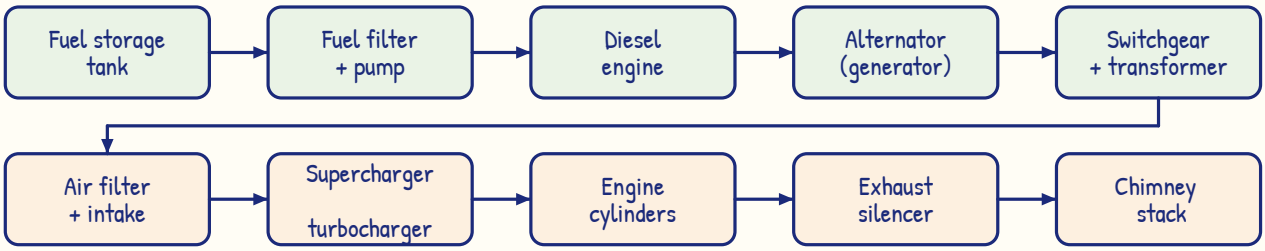


Fig: Block diagram of a diesel power plant

Main parts

- **Diesel engine** (prime mover) coupled directly to an **alternator** (synchronous), usually 3-phase, 415 V / 11 kV, 1500 rpm (4 poles) or 750 rpm.
- Foundations: heavy, **vibration-isolated** concrete block with a gap to the building floor.
- **Control room** with switchgear, **governor** (maintains constant speed), AVR (voltage regulator), protection panels.

Auxiliary systems

System	Function
Fuel supply system	Storage tank (bulk) -> transfer pump -> day tank -> filter -> **fuel injection pump** -> injector (atomiser, 150-300 bar)
Air intake system	Air filter (dry / oil bath), intake manifold, **supercharger / turbocharger** to raise power by 30-50%
Exhaust system	Exhaust pipe, **silencer / muffler**, waste heat boiler, chimney; reduces noise and carries gases outside
Cooling system	Jacket water around cylinder liners (pump, **radiator or cooling tower**), keeps temp ~ 70-85 deg C
Lubrication system	Oil sump, pump, filter, cooler; reduces friction and wear and cools the piston; **wet sump / dry sump**
Starting system	Electric starter motor and battery (small); **compressed air** at 20-30 bar (large)
Governing system	Mechanical / electronic governor controls fuel to hold the speed against load changes

Fuel storage

- Diesel (HSD, LDO, heavy fuel oil) is stored in **underground or above ground steel tanks**, with enough for **7-15 days** run; a smaller **day tank** (about 8 hours) is kept near the engine and fed by gravity.
- Precautions: fire safety, tank vent, **dike (bund wall)** for leakage, settling of water and sludge (drain cock), heating of heavy oils for pumping, fuel quality (**cetane number 45-55**, viscosity, sulphur, flash point).
- Fuel **filters** (primary + secondary) are important as injector clearances are tiny.

Fuel injection

- **Air injection** (obsolete) and **solid injection**: **common-rail**, unit injector, pump-line-nozzle systems. A good spray needs: atomisation, penetration, uniform distribution.
* Plant efficiency (overall) 30–38%. Heat balance of a diesel engine (approx.): useful work 35%, exhaust gas 30%, jacket cooling water 30%, friction and radiation 5%.

Waste heat

- The exhaust (350–450 deg C) is passed through a **waste-heat boiler** to raise steam or hot water -> improves overall utilisation (**cogeneration**, 60–70%).

Supercharging and governing

- **Supercharging** raises inlet air pressure (1.2–2.5 bar) so more air (oxygen) enters -> more fuel burnt -> **power up 30–50%**. Types: **turbocharger** (exhaust-gas driven turbine, most common), mechanical (gear-driven blower), with an **intercooler / after-cooler** for denser air.
- **Governing:** quantity governing (change fuel quantity per cycle, **used in diesel**) -- as the load rises speed falls, the governor increases fuel to restore rated speed. Parallel operation needs governor **droop 3–5%** for load sharing.

Cooling and lubrication in detail

- **Cooling:** about 30% of heat goes to the jacket water. Open (thermosyphon), closed with **radiator + fan**, or cooling-tower system; coolant treated with corrosion inhibitors; thermostat keeps 70–85 deg C. Too cold -> corrosion, poor combustion; too hot -> seizure, oil breakdown.
- **Lubrication:** reduces friction (**30% of mechanical loss**), cools, seals the piston rings, cleans, and protects against corrosion. Oil pressure 3–5 bar; filter, cooler and centrifuge keep it clean.
- **Starting:** electric motor (up to ~ 100 kW), compressed air (valve opens air into cylinders in firing order), or **air motor**. Black-start is possible, important in grid failure.

Selection of site and size

- Near the **load centre**, fuel supply route, small water source, away from residential areas (noise) with room for tank farm; foundations on hard soil.
- Number of units: several **equal-size sets** (n + 1 with one spare) give better part-load efficiency and reliability.

Typical Indian applications

- Remote / island electrification, **telecom towers**, hospitals and data centres (UPS + DG), construction sites, **captive power** in industries, **peak** shaving, and as **black-start units** for big power stations.

4. Performance and Examples of Diesel Plants

Example 1: Diesel set

- A diesel engine uses 800 kg of fuel per hour (CV = 42,000 kJ/kg) and delivers 3200 kW of brake power.
- Brake thermal efficiency = 3200 x 3600 / (800 x 42,000) = 11.52×10^6 / 33.6×10^6 = **0.343 = 34.3%**
- sfc = 800 / 3200 = **0.25 kg / kWh**

Example 2: Cost per unit

- 2 MW set, load factor 50%: energy per year = 2000 x 0.5 x 8760 = **8.76×10^6 kWh**; diesel at Rs 90 / kg with sfc 0.25 -> fuel cost = **Rs 22.5 / kWh**; add maintenance and fixed charges -> Rs 25-30 per kWh -- expensive, hence used for stand-by and peaking only.

5. Gas Turbine Plant - Introduction and Classification

- **Gas turbine (GT) plant** uses hot, high-pressure **gas** (products of combustion) to rotate a turbine and drive an alternator. It works on the **Brayton (Joule) cycle**.
- Basic parts: **compressor, combustion chamber, turbine** (+ starting motor, controls, auxiliaries).

Classification

- **By cycle:** (1) **Open cycle** -- air taken from atmosphere, exhaust released. (2) **Closed cycle** -- the same working fluid (air, helium) circulates; heat is added externally. (3) Semi-closed.
- **By combustion:** constant-pressure (**most**) and constant-volume (explosion - obsolete) type.
- **By use:** industrial / power (heavy-duty frame, e.g. 100-400 MW class), **aero-derivative** (light, quick), micro-turbines (< 500 kW), marine, and locomotive.
- **By compressor:** axial flow (large) or centrifugal (small).

6. Construction of Gas Turbine Plant

- **Compressor:** axial-flow multi-stage (pressure ratio 10-30), driven by the turbine itself (**compressor takes ~ 50-60% of the turbine work**). **Centrifugal** for small ones.
- **Combustion chamber:** can, annular or can-annular; air divided into **primary** (burns fuel), **secondary** (completes combustion) and **tertiary / dilution air** (cools gas to turbine inlet temperature of **1100-1500 deg C**). Fuel: natural gas, diesel, kerosene.
- **Turbine:** axial flow, 2-4 stages, blades from nickel super-alloys with **film cooling** and thermal barrier coating. Speed 3000 / 3600 rpm.
- **Auxiliaries:** starting motor / diesel engine, **lubricating oil system**, fuel system, air filter, **inlet air cooler / fogging**, silencer, **governing** (fuel control), generator, exhaust diffuser + stack, and the **regenerator / heat exchanger** (if fitted).

7. Layout of Gas Turbine Plant with Auxiliaries and Working Principle

Open cycle gas turbine

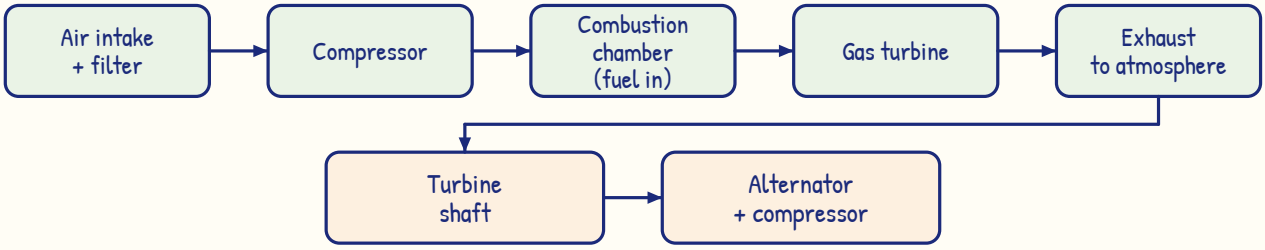


Fig: Open-cycle gas turbine plant

- Air is compressed (1-2), heated by burning fuel at constant pressure (2-3), expanded in the turbine (3-4) and exhausted. **Net work** = turbine work - compressor work.

Closed cycle gas turbine

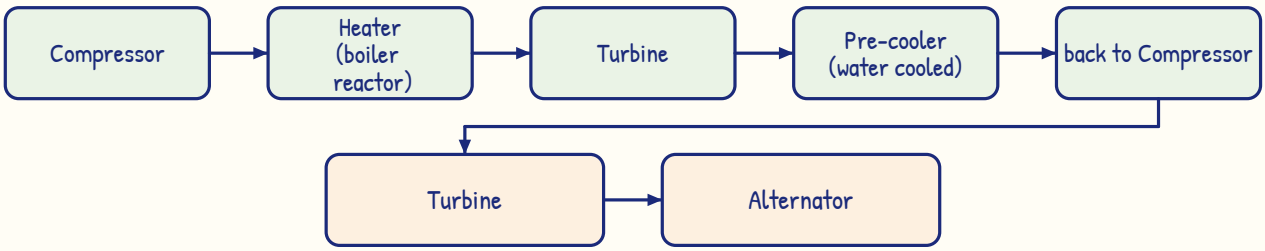


Fig: Closed-cycle gas turbine plant

- Working fluid (air, helium, CO_2) circulates in a closed loop; heat is supplied in a **heater** from external fuel, even **coal or nuclear** (high-temperature gas-cooled reactor); **pre-cooler** rejects heat. Can use **high pressure** (smaller size, better heat transfer).

8. Brayton Cycle Analysis

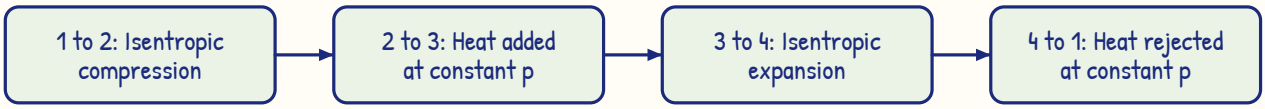


Fig: Ideal Brayton cycle processes

Formulae (ideal air standard)

- Pressure ratio $r_p = p_2 / p_1$; $T_2 / T_1 = r_p^{(\text{gamma}-1)/\text{gamma}}$; T_3 / T_4 = same
- Compressor work $W_C = c_p (T_2 - T_1)$; Turbine work $W_T = c_p (T_3 - T_4)$; Net work $W = W_T - W_C$

- Heat added $Q_1 = c_p (T_3 - T_2)$. **Thermal efficiency = 1 - 1 / $r_p^{(gamma-1)/gamma}$**
- **Work ratio** = W_{net} / W_T (0.3–0.6 for GT; bad if compressor / turbine efficiencies are low)
- Optimum pressure ratio for **max. work**: $r_p = (T_3 / T_1)^{gamma / (2(gamma-1))}$

- Efficiency depends only on pressure ratio. Specific work depends on **turbine inlet temperature (TIT)**. Real cycle: isentropic efficiencies of compressor eta_c (0.85) and turbine eta_t (0.88).

Methods to improve performance

- **Regeneration (heat exchanger):** exhaust heat preheats compressed air -> saves fuel; effectiveness 0.6–0.8; useful at low pressure ratio.
- **Intercooling:** cool air between compressor stages -> less compressor work.
- **Reheating:** gas reheated between turbine stages -> more turbine work.
- Use of all three approaches Ericsson cycle; but adds complexity and pressure drop.

Example 3: Brayton cycle

- Air at 1 bar, 300 K compressed with $r_p = 8$, turbine inlet 1200 K. gamma = 1.4, c_p = 1.005 kJ/kg K.
- $T_2 = 300 \times 8^{0.2857}$ = **543.4 K**; T_4 = 1200 / 1.811 = **662.6 K**
- W_C = 1.005 (543.4 - 300) = **244.6 kJ/kg**; W_T = 1.005 (1200 - 662.6) = **540.1 kJ/kg**; W_{net} = **295.5 kJ/kg**
- Efficiency = 1 - 1/1.811 = **0.448 (44.8%)**; Work ratio = 295.5 / 540.1 = **0.547**

Example 3b: Regenerative Brayton cycle

- Same data as Example 3 (T_2 = 543.4 K, T_4 = 662.6 K) with a regenerator of **effectiveness 0.7**.
- Air leaving regenerator $T_x = T_2 + 0.7 (T_4 - T_2)$ = 543.4 + 0.7 x 119.2 = **626.8 K**

- Heat added = 1.005 (1200 - 626.8) = **576.1 kJ/kg** (earlier 659.9); Efficiency = 295.5 / 576.1 = **0.513 (51.3%)** -- better than 44.8%.

Aero-derivative and heavy-duty turbines

- **Heavy-duty (frame) GT:** thick casing, long life, 50-500 MW, lower efficiency in simple cycle (33-40%), used in CCGT. Examples of frame classes: E, F, H, J.
- **Aero-derivative GT:** based on jet engines; light, compact, **start in 2-5 min**, 20-100 MW, efficiency 35-42%; for peaking, offshore, ships.
- Typical TIT: older 900 deg C -> F-class 1400 deg C -> H/J class **1600 deg C** (advanced cooling + coatings); **ISO rating** = 15 deg C, 1.013 bar, 60% RH.

Effect of ambient conditions

- Output drops about **0.7-1% for every 1 deg C rise** in ambient temperature -> use **inlet air chillers / evaporative cooling / fogging**.
- Efficiency also reduced by altitude and by **compressor fouling** (water wash), inlet and exhaust pressure drops.

Fuels for GT

- Natural gas (preferred, clean), naphtha, kerosene, HSD, syngas (IGCC), biogas, and **hydrogen blends**. Liquid fuel needs good atomisation and clean (no vanadium / sodium) to avoid hot corrosion of blades.

Gas turbine governing and control

- Fuel flow is varied to hold speed (frequency); controls: **temperature limit** (TIT), acceleration limit, compressor **surge protection** (bleed valves, variable inlet guide vanes).
- **Starting sequence:** purge -> crank by starting motor (or static frequency converter) to ~ 20% speed -> ignite -> accelerate -> self-sustaining -> synchronise.

9. Advantages and Disadvantages of Gas Turbine Plants

Advantages	Disadvantages
Small size, light weight, simple layout	Low efficiency (open cycle simple: 25-35%)
Quick start (5-10 min); good for peak load	Compressor takes large part of turbine work
Little cooling water required	High-grade fuel (gas, distillate) is costly
Low capital cost per kW and short installation time	Blades face very high temperature -> special alloys
Smooth running, less vibration, low lubrication oil use	Short life, noise (intake + exhaust), part-load efficiency poor
Wide fuel flexibility, easy automatic operation	Output falls on hot days (air less dense) and at altitude

Point	Diesel plant	Gas turbine plant	Steam plant
Size (MW)	up to 50	up to 400	up to 1000+
Start-up	1-2 min	5-10 min	hours
Efficiency	35-40%	30-40%	35-45%
Water needed	low	very low	very high
Fuel	HSD / HFO	gas / distillate	coal etc.
Use	stand-by, peak	peak, combined cycle	base load

10. Combined Cycle Power Plants

- ***Combined cycle (CCGT)*** = a ***gas turbine (Brayton, topping cycle)*** + a ***steam turbine (Rankine, bottoming cycle)***. The hot exhaust (500-600 deg C) of the GT goes to a ***heat recovery steam generator (HRSG)*** to raise steam.

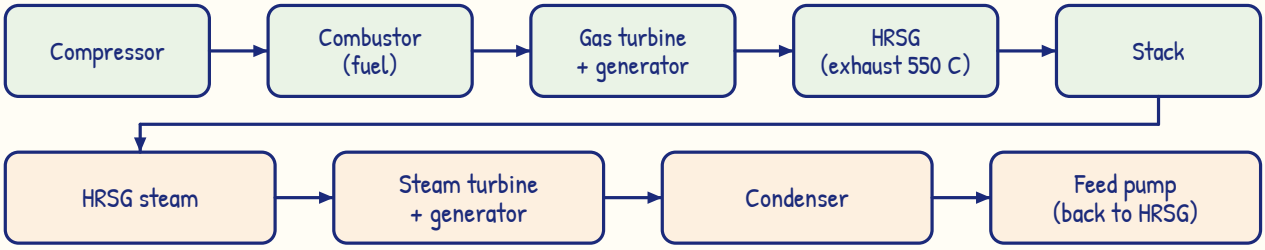


Fig: Combined-cycle power plant

- **Efficiency:** GT 35-40% + ST contribution -> overall **55-62%** (best among thermal plants); about **two-thirds of power from GT** and one-third from ST.
- Plant sizes: 100-1000 MW; unit uses 1 GT + 1 ST (single shaft) or 2 GT + 1 ST (**2 x 1 multi-shaft**).

HRSG

- Parts: **superheater, evaporator, economiser** (counter-flow), drum, **pinch point** temperature difference 10-20 deg C. Types: single, double and triple-**pressure** (HP, IP, LP) with reheater to raise efficiency. Supplementary firing (duct burners) can raise steam output.

Combined-cycle efficiency formula

- $\eta_{cc} = \eta_{GT} + \eta_{ST}(1 - \eta_{GT})$ - losses. Example: $0.36 + 0.33(1 - 0.36) = 0.36 + 0.211 =$ **0.571 = 57%**

Advantages

- **Highest thermal efficiency**, low fuel use, **low CO_2 per kWh** (about 40-50% less than coal), very low SO_x, lower water need than steam plant, quick installation (**2 years**), flexible, modular, low capital cost / kW, can be started in 30-60 minutes.

Disadvantages

- Needs clean **gas fuel** (gas supply is uncertain in India), efficiency drops at part load and on high ambient temperature, complex controls, NO_x control needs **DLN burners** or water / steam injection.
- Other forms: **IGCC** (coal gasification + combined cycle), **Integrated solar combined cycle**, cogeneration with process steam.

Applications of GT and CCGT

- Peak power, base load (CCGT) where gas is available, **gas pipelines**, offshore platforms, aircraft / ship propulsion, **cogeneration** in refineries / paper / fertiliser, power supply on trains and emergency standby.

II. Important Short Questions (exam revision)

Q & A

- **Why is a diesel plant used as stand-by?** Quick start, no stand-by loss, small size; but running cost is high.
- **What is the compression ratio of a diesel engine?** 14:1 to 22:1 (higher than petrol engine, as only air is compressed).
- **Why is a silencer used?** Exhaust gas leaves at high pressure and velocity; silencer reduces noise and pulsations.
- **Why is a governor needed?** To keep speed (and so frequency, 50 Hz) constant by varying the fuel supplied as the load changes.
- **What is surge in a GT compressor?** Reverse flow at low mass flow / high pressure ratio causing violent vibration; prevented by bleed valves.
- **Why does a GT compressor consume 50-60% of turbine work?** Gas is compressed in a cold state; gases expand hot with a large volume, so turbine produces much more work.
- **Topping and bottoming cycle?** Topping (Brayton, high temperature) rejects heat at ~ 550 deg C to the bottoming (Rankine, low temperature).
- **What is the pinch point in an HRSG?** Minimum temperature difference between exhaust gas and water / steam at the evaporator inlet; small value gives more steam but bigger area.
- **Closed cycle advantage?** Clean working fluid (no blade fouling), any fuel, high pressure -> compact; but needs a heat exchanger (large) and cooling water.

Quick revision

- Diesel plant: quick start, 35-40% efficiency, stand-by / peak use. 4-stroke vs 2-stroke. Systems: fuel, air, exhaust, cooling, lubrication, starting.
- Brayton efficiency = $1 - 1 / r_p^{(gamma-1)/gamma}$. Improve by regeneration, intercooling, reheating.
- Open cycle: air + fuel, exhaust to atmosphere. Closed cycle: working fluid recirculated, external heater.
- Combined cycle = GT + HRSG + ST, efficiency 55-60%.

POWER PLANT ENGINEERING

Subject Code: 23A03702e

UNIT IV: Hydro Electric Power Plant

1. Hydro Electric Power Plant – Water Power

- ***Hydro power*** converts the ***potential energy of water stored at a height*** (head) into kinetic energy -> mechanical energy in a ***water turbine*** -> electrical energy in a generator.

> *Power P = rho g Q H eta (watts) = 9.81 Q H eta (kW), with Q in m^3/s and H (net head) in metres*

- Example: Q = 20 m^3/s, H = 100 m, overall eta = 0.85 -> P = 9.81 x 20 x 100 x 0.85 = ***16,677 kW = 16.7 MW***
- Energy from hydro is ***renewable, clean, cheap to run*** (no fuel), long life (50–100 years), quick start (2–5 min), and gives ***flood control, irrigation, drinking water, navigation and fisheries***.

Advantages / disadvantages

Advantages	Disadvantages
No fuel cost, low running + maintenance cost	***High capital cost***, long construction time
No pollution; no ash or smoke	Depends on rainfall -> output varies (dry years)
Quick start-up and load pickup (peak + spinning reserve)	Submergence of land, forests, villages (***rehabilitation***)
Long life, high efficiency (85–90%)	Sites are far from the load centre -> long transmission lines
Multi-purpose benefits (irrigation, flood control)	Silting of reservoir; ecological change; earthquakes risk

2. Hydrological Cycle

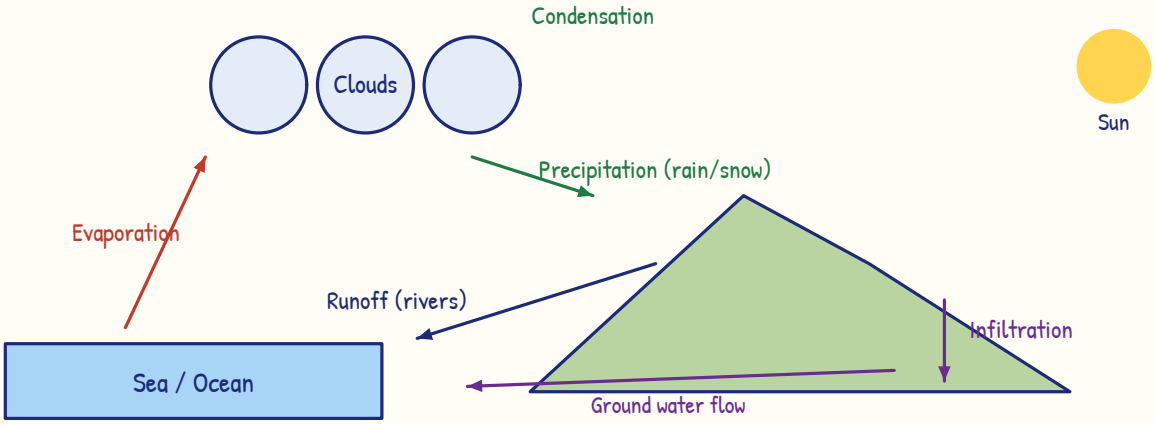


Fig: Hydrological (water) cycle

- Sun's heat -> ***evaporation*** from oceans / lakes / soil, ***transpiration*** from plants -> vapour rises -> ***condensation*** into clouds -> ***precipitation*** (rain, snow, hail) -> ***runoff*** in rivers, ***infiltration*** into ground (ground water) -> back to sea.
- Rainfall in India mainly from the ***south-west monsoon (June – Sept.)*** -- average ~ 1170 mm.

Precipitation measurement

- ***Rain gauge:*** non-recording (Symon's, cylindrical collector) and recording (***tipping bucket, weighing, float type***). The ***average*** rainfall over a basin from many gauges: ***arithmetic mean, Thiessen polygon, isohyetal method*** (most accurate).
- Runoff = Precipitation – Losses (evaporation, infiltration, interception, depression storage). ***Runoff coefficient*** = runoff / rainfall (0.1–0.9; depends on slope, soil, vegetation).

3. Flow Measurement – Streamflow

- ***Rate of flow (discharge) Q = A x V*** (m^3/s). It is needed to find the hydro potential and to design the plant.

Method	Principle
Velocity-area method	Cross-section divided into strips; velocity in each measured by a ***current meter*** (cup / propeller type) at 0.6 depth or 0.2 + 0.8 depth average; Q = sum of (area x velocity)

Method	Principle
Float method	Surface float travels a known distance in time t; mean velocity = 0.85 x surface velocity (approx.)
Weirs and notches	$Q = (2/3)\ C_d\ L\ \sqrt{2g}\ H^{1.5}$ for rectangular weir (small streams); V-notch: $Q = (8/15)\ C_d \tan(\theta/2)\ \sqrt{2g}\ H^{2.5}$
Chemical / salt dilution	Tracer injected, concentration downstream measured; for turbulent mountain streams
Stage-discharge (rating curve)	Gauge height versus measured Q plotted; discharge read from stage thereafter
Ultrasonic / electromagnetic	Modern, continuous measurement

- **Chezy / Manning formula** for uniform flow: $V = (1/n)\ R^{2/3}\ S^{1/2}$ (R = hydraulic radius, S = slope, n = roughness).

4. Drainage Area (Catchment) Characteristics

- **Catchment / drainage basin** = the area from which rainfall drains to a given point of a river. Its **shape, size, slope, soil, vegetation and geology** decide runoff.
- **Size:** larger area -> more runoff, slower response. **Shape:** fan-shaped (round) gives a sharp, high flood peak; long, narrow shape gives a low and flat hydrograph.
- **Slope:** steep -> rapid runoff, less infiltration. **Soil:** impervious (clay, rock) -> high runoff; sandy -> high infiltration.
- **Vegetation / land use:** forests reduce and delay runoff, reduce erosion; urbanisation increases runoff.
- **Drainage density** = total length of streams / area (high density -> fast runoff). **Stream order** (Horton's) and **river slope** are also considered.
- **Evaporation and weather:** rainfall intensity, duration, distribution, temperature, wind.

5. Hydrographs

- A **hydrograph** = plot of **discharge versus time** at a point of a river (stream).

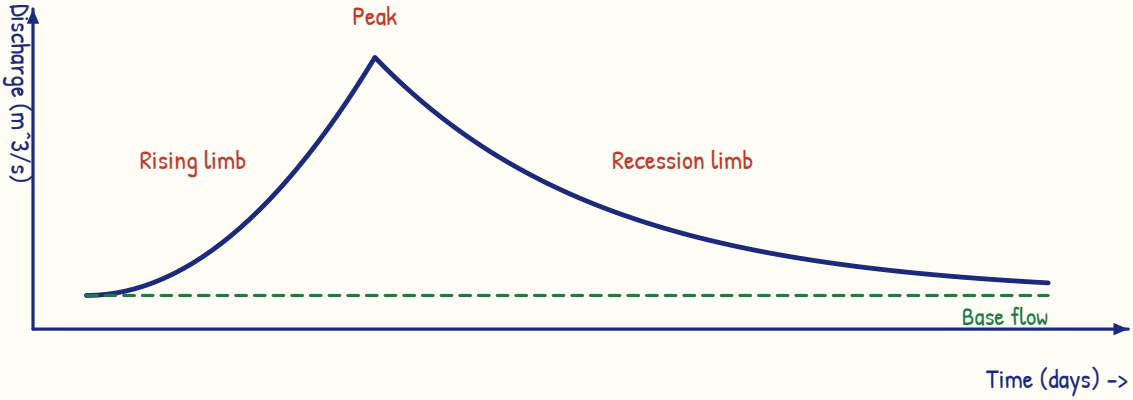


Fig: Flood hydrograph with rising limb, peak and recession limb

- Parts: **rising limb** (surface runoff builds), **crest / peak**, **recession (falling) limb**, and **base flow** (ground water contribution).
- **Unit hydrograph:** direct runoff hydrograph from **1 cm of effective rainfall** of a given duration (D hours) spread uniformly over the basin. Used (linear superposition) to predict floods from a storm.
- **Hyetograph:** rainfall intensity versus time (bar graph).

Flow duration curve

- Plot of discharge (Q) against the **percentage of time** that this discharge is equalled or exceeded. The area under the curve gives the volume of water / **power potential**. Used for choosing plant capacity, **firm (dependable) power** (e.g. Q_{90} or Q_{75}), and the type of plant (run-of-river or storage).

Mass curve (Ripl's method)

- Plot of **cumulative inflow volume versus time**. The slope = flow rate. A line of constant demand drawn tangent to the crests gives the **required reservoir capacity** (maximum vertical distance between the demand line and the mass curve).

Example 1: Reservoir capacity

- Mass curve for the lean months: inflow 40, 30, 20, 25 Mm^3; steady demand 35 Mm^3 each month.
- Deficit: 0, 5, 15, 10 -> cumulative deficit 0, 5, 20, 30 Mm^3. Reservoir capacity required = **30 Mm^3**.

6. Storage and Pondage

- **Storage:** large reservoir that holds **excess flow of the rainy season** to be used in the dry season (seasonal regulation). Plants with storage are mostly **peak or base** load plants and give a steady output.
- **Pondage:** small storage behind a **run-of-river** plant weir / barrage which stores water during **off-peak hours of the day (night)** to meet the **peak demand** (daily / weekly regulation).
- **Plant use factor** = actual energy generated / (installed capacity x hours of operation). Pondage allows a run-of-river plant to run at higher capacity during peak hours.
- **Reservoir zones:** dead storage (silt), live (useful) storage, flood control storage (above full reservoir level).

Level	Meaning
Minimum Drawdown Level (MDDL)	Lowest level for power; below this dead storage
Full Reservoir Level (FRL) / Normal Pool	Maximum level for normal storage
Maximum Water Level (MWL)	Highest level during design floods (with spillway)
Free board	Height of dam top above MWL (waves, safety)

- **Reservoir sedimentation:** silt reduces capacity (0.5–1% per year); controlled by catchment treatment, silt sluices, dredging, trap efficiency, bypass.

7. Classification of Dams

- **Dam** = a barrier built across a river to create a reservoir and **head**. Choose by site, foundation, material, climate, cost.

Type	Features
Gravity dam	Masonry / concrete; stability by its **own weight**; needs good rock foundation; most common for 50+ m; Bhakra, Nagarjuna Sagar (masonry)

Type	Features
Arch dam	Curved upstream in plan; transmits water load to abutments by arch action; narrow gorge, strong rock sides; less concrete; Idukki
Buttress dam	Slanting deck supported by buttresses; saves concrete; wide valley; skilled labour
Embankment dam (earth / rock-fill)	Earth or rock; good for weak foundations; cheaper; needs a **spillway** separate; Tehri (earth + rock-fill)
Composite dam	Concrete + embankment parts

Forces on a gravity dam

- Water pressure (horizontal), **self weight**, **uplift pressure**, silt pressure, wave, ice, earthquake, wind.
- Checks: no **overturning**, **sliding** (factor of safety > 1.5 with shear friction), no **tension** in concrete, compressive stress within limit; resultant must lie in the **middle third** of the base.

8. Spillways

- A **spillway** safely discharges surplus / **flood** water from the reservoir to the downstream river **without damaging the dam**.

Spillway	Description
Overflow (ogee)	Crest shaped like the lower nappe of a jet over a sharp weir; on **concrete dams**; with gates
Chute (trough / open channel)	Steep open channel to the side of the earth dam
Side-channel	Flow goes over crest, then turns 90 deg in a side channel parallel to the crest
Shaft (morning glory)	Funnel-shaped inlet, vertical shaft, then tunnel; for narrow gorges
Siphon	Automatic; primes and discharges at the full level; for small variation

Spillway	Description
Tunnel / Conduit	Water carried through hill or dam body

- **Gates:** vertical lift, **radial (Tainter) gates** (most used), drum, rolling. **Energy dissipators** (stilling basin, flip bucket, hydraulic jump) at the toe to reduce scour.
- Design flood: standard project flood (SPF) / **probable maximum flood (PMF)** based on rainfall.

9. Hydro Power Plant – Types and Classification

- **Based on water use / availability:**
  - **Run-of-river:** no or little storage; output varies with flow; low head; base plants on rivers; small / mini / micro hydro.
  - **Storage (reservoir):** large dam; steady operation; for base, peak loads.
  - **Pumped storage:** water pumped up at off-peak and reused at peak.
  - **Tidal:** see Unit V.
- **Based on head: Low (< 15 m), medium (15–70 m), high (> 70 m)** head plants.
- **Based on load:** base load, peak load plants.
- **Based on capacity:** pico (< 5 kW), micro (< 100 kW), mini (< 2 MW), small (< 25 MW), large (> 25 MW).
- **Based on location of power house: surface, semi-underground** and **underground** power house; **dam-toe, canal-side** and **conduit** type.

Turbine selection (summary)

Turbine	Type	Head	Specific speed
Pelton wheel	Impulse	High (> 250 m), small Q	10–35
Francis	Reaction (radial / mixed)	Medium (40–400 m)	60–300
Kaplan / propeller	Reaction (axial)	Low (< 40 m), large Q	300–1000

- Pelton: jets strike **buckets**; nozzle with spear controls flow. Francis: spiral casing, guide vanes, draft tube. **Kaplan:** adjustable runner blades -> good part-load efficiency.

Water turbines in more detail

- **Pelton wheel (impulse):** all head is converted to kinetic energy in the nozzle; jet velocity $V_1 = C_v \sqrt{2gH}$ (C_v = 0.97–0.99); the wheel works at **atmospheric pressure**; bucket speed $u = 0.45 \times V_1$ (ideal **0.5 V_1**); runs 12–35 buckets; needs **no draft tube**; efficiency 85–90%. Governing by **spear (needle) valve** and **deflector** (to avoid water hammer).
- **Francis (reaction):** water enters through **spiral casing** -> **stay vanes** -> adjustable **guide vanes** -> runner (radial inward, mixed-flow exit) -> **draft tube**. Part of the head is pressure head. Efficiency up to 93–95%. Governing by guide vane opening.
- **Kaplan (axial reaction):** runner like a ship propeller with **adjustable blades** (and adjustable guide vanes: **double regulation**) -> high efficiency even at part load; for large discharge, low head (bulb turbines for 2–15 m heads).

> Specific speed $N_s = N \sqrt{P} / H^{5/4}$ (N in rpm, P in kW, H in m)

- **Unit quantities:** $N_u = N / \sqrt{H}$; $Q_u = Q / \sqrt{H}$; $P_u = P / H^{3/2}$ -- used to predict the performance at a different head (**similarity / model laws**).
- **Cavitation:** formation and collapse of vapour bubbles where local pressure falls below vapour pressure -> pitting, noise, vibration, fall in efficiency. Controlled by limiting suction (draft) head, proper runner design, use of **stainless steel**, and the **Thoma cavitation factor** $\sigma = (H_{atm} - H_{vap} - H_s) / H$.
- **Draft tube:** diverging tube (conical / elbow) which **recovers kinetic energy** and lets the runner be above tail-water level; efficiency 70–90%.

Example 3: Pelton wheel power

- A jet of water 0.1 m^3/s under a net head of 400 m strikes a Pelton wheel (C_v = 0.98, overall efficiency 0.85).
- Jet velocity = $0.98 \sqrt{2 \times 9.81 \times 400} = 0.98 \times 88.6$ = **86.8 m/s**
- Shaft power = $9.81 \times 0.1 \times 400 \times 0.85$ = **333.5 kW**

Governing of hydro turbines

- Mechanical-hydraulic or electro-hydraulic **governor** with speed sensor (flyball / tachometer), servo-motor and oil pressure unit. Speed regulation 3–5%.
- **Water hammer** = pressure rise when flow is stopped suddenly (Joukowsky): delta p = rho c V; closing time of guide vanes is chosen long enough (> 2L/c) and a **surge tank** or **pressure relief valve** is provided.

Power estimation with flow duration

Example 4: Energy from a run-of-river site

- Available head 30 m, plant efficiency 0.85. A flow duration curve gives: 40 m^3/s for 30% of year, 25 m^3/s for 40%, 10 m^3/s for 30%.
- Power: 10,006 kW, 6254 kW and 2502 kW respectively; Energy per year = 8760 x (0.3 x 10,006 + 0.4 x 6254 + 0.3 x 2502) = **8760 x 6254 = 54.8 x 10^6 kWh**
- Average load = 6254 kW; with plant of 10 MW -> **capacity factor 62.5%**; firm power ~ 2.5 MW.

10. Typical Layouts and Plant Auxiliaries

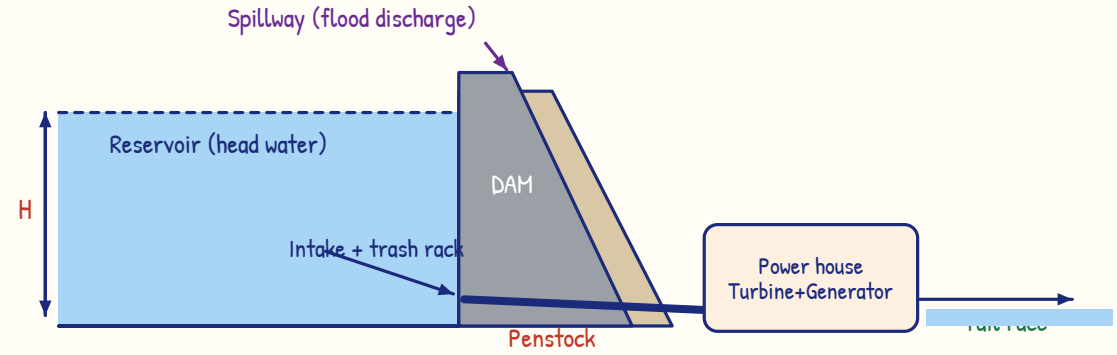


Fig: Layout of a storage-type hydro plant (dam, penstock, power house)

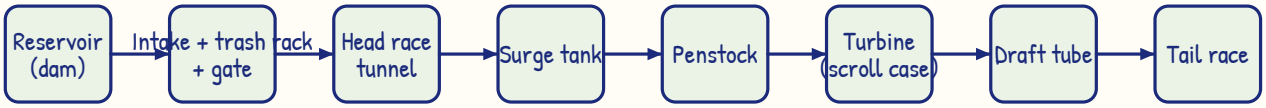


Fig: Components in the water path

- **Dam and reservoir** hold water and give head. **Intake / trash rack** stops debris; gates control flow.

- **Penstock:** steel / RCC pipe carrying water at high pressure to the turbine; **water hammer** is controlled by the surge tank and relief valves.
- **Surge tank:** an open chamber connected to the penstock which **absorbs the pressure surge** from sudden load change (reduces water hammer) and provides water during rapid load rise. Types: simple, restricted orifice, differential (Johnson), closed.
- **Power house:** houses turbines, generators, control room, **EOT crane**, switchyard. **Draft tube** recovers the exit kinetic energy and allows the turbine to be above tail water. **Tail race** returns the water to the river.

Plant auxiliaries

- **Governor** (oil pressure actuated; changes gate / needle opening to hold speed), **inlet valve (butterfly / spherical)**, **cooling water system** for generator and bearings, **lubrication** and **brake system**.
- **Excitation system**, transformers, switchyard (circuit breakers, isolators, lightning arresters), **DC battery** supply, fire protection, drainage and dewatering pumps, ventilation (for underground plants), cranes.
- **Trash-rack cleaning machine, fish ladder / pass**, cathodic protection, instrumentation and SCADA.

II. Plant Operation

- **Start-up:** open inlet valve bypass -> fill the penstock and scroll case -> open guide vanes -> accelerate to rated speed -> excite generator -> **synchronise** with grid -> load by opening the guide vanes.
- **Shut-down:** unload, trip the breaker, close guide vanes, close inlet valve (and the intake gate for maintenance); brakes applied at low speed (< 30%).
- **Control:** The **governor** keeps speed (frequency) constant. **Load-frequency control (AGC)** and **reactive power / voltage control** by the AVR.
- **Operation in the grid:** hydro units pick up load in a few minutes -> used as **peaking stations** and **spinning reserve**, **frequency regulation**; thermal / nuclear take the base load.
- **Operating rules:** avoid rough zones (vibration, cavitation), monitor bearing temperatures, vibration, air gap, oil levels; **cavitation** damage at runner blades is prevented by correct setting of turbine and by using stainless steel.

- Maintenance: annual overhaul in dry period; silt erosion repair; inspection of dam (***dam safety***) and instruments (piezometers, seepage, tilt meters).

12. Pumped Storage Plants

- Used with thermal / nuclear base-load plants. ***Off-peak*** surplus power (night) is used to ***pump water from a lower reservoir to an upper reservoir***; during ***peak*** hours the water flows down through turbines to generate power.

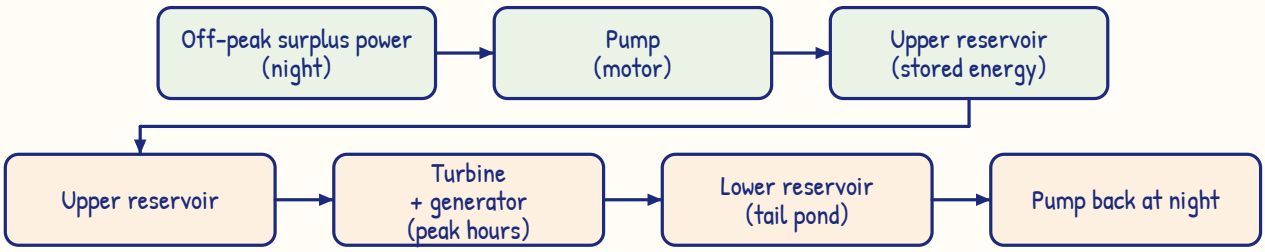


Fig: Working of pumped storage plant

- Arrangements: ***two-reservoir*** type (separate pump and turbine) or ***reversible pump-turbine*** with motor-generator (***most economical***; Francis type reversible).
- ***Overall (cycle) efficiency 70–80%*** (pump 90% x turbine 90% x losses). It gives ***load levelling***, improves system load factor, and acts as a ***huge battery***; useful with ***solar / wind*** integration.
- India: Kadamparai, Bhira, Srisailam LBPH, ***Purulia (900 MW)***, Tehri PSP, Kadana. Potential about 96 GW.

Example 2: Pumped storage

- Upper reservoir 100 m above lower; volume 5×10^6 m^3. Energy stored = rho g V H = $1000 \times 9.81 \times 5 \times 10^6 \times 100 = 4.9 \times 10^{12}$ J = ***1362 MWh***.
- With round-trip efficiency 75%, recoverable energy = ***~ 1020 MWh***

13. Hydro Power in India and Small Hydro

- ***Major stations:*** Bhakra-Nangal (1325 MW, Gobind Sagar), Tehri (1000 MW + 1000 MW PSP), ***Sardar Sarovar***, Koyna (Maharashtra), Nagarjuna Sagar, Srisailam, Idukki (arch dam), Sharavathi, Subansiri, Kishanganga.
- Share of hydro in installed capacity ~ ***11%*** (earlier ~ 40–45%); it is important as ***peaking*** and for ancillary services. Development agencies: NHPC, SJVNL,

state boards, THDC.

- **Small hydro (SHP):** up to **25 MW**; mini (100 kW – 2 MW), micro (up to 100 kW); simple run-of-river, **no large dam**, little displacement; suitable for hilly, remote areas; classes by head: ultra low (< 3 m), low (3–30), medium, high.
- Pico / micro hydro: **water mills (gharats)** modernised, crossflow (**Banki**) and Turgo turbines, induction generators; cost Rs 1–1.5 lakh per kW.

Hydro-thermal co-ordination

- Hydro plants take **peak load** and thermal / nuclear plants run at constant base load -> system operates economically.
- In a **wet season** run hydro as base to use the water that otherwise spills; in a **dry season** keep hydro for peak hours only.
- Hydro units are **excellent for frequency control** because of fast loading rate (**~ 20–50% of capacity per minute**).

14. Short Questions (exam revision)

Q & A

- **Why is a surge tank provided?** To relieve water hammer pressure in the penstock and to supply extra water when the load increases suddenly.
- **Difference between storage and pondage?** Storage = seasonal, large reservoir; pondage = small, daily / weekly regulation for run-of-river plants.
- **What is the unit hydrograph?** Hydrograph of direct runoff resulting from unit (1 cm) depth of effective rainfall uniformly spread on the basin in specified duration.
- **Why Kaplan for low head?** Large flow requires large flow area; axial flow runner with adjustable blades gives high efficiency for large Q at low head.
- **What is a draft tube?** Gradually expanding tube connecting the runner exit to tail race; converts kinetic energy to pressure, raising the effective head.
- **What is the firm power?** Power continuously available from a plant at all times (~ 90–95% of the time).

- **Run-of-river advantage?** Low cost, no submergence; drawback: output varies with river flow.

Quick revision

- P (kW) = 9.81 Q H eta. Pelton: high head; Francis: medium; Kaplan: low head.
- Storage = seasonal; pondage = daily. Surge tank = relieves water hammer. Spillway = flood escape.
- Pumped storage: pump at night, generate at peak; efficiency 70–80%.

POWER PLANT ENGINEERING

Subject Code: 23A03702e

UNIT V: Non-Conventional Power and Nuclear Power Station

1. Solar Energy – Introduction

- The Sun is a gaseous sphere (diameter 1.39 x 10^6 km, surface temp. ~ 5800 K); energy from ***nuclear fusion*** of hydrogen into helium.
- ***Solar constant*** I_{sc} = ***1367 W/m²*** (energy per unit time on unit area perpendicular to rays, at mean earth-sun distance, outside the atmosphere). About ***1000 W/m²*** reaches the ground on a clear day.
- Types of radiation: ***beam (direct)***, ***diffuse*** (scattered by atmosphere) and ***global (total) = beam + diffuse***; reflected (albedo).
- India receives ***4–7 kWh/m²/day*** with 250–300 clear days -> a huge potential (~ 750 GW). Instruments: ***pyranometer*** (global / diffuse), ***pyrheliometer*** (beam), sunshine recorder (Campbell-Stokes).

Utilisation of solar energy

- ***Thermal route:*** solar water heater, dryer, cooker, still, ***solar thermal power*** (concentrating), space heating, refrigeration / air conditioning.
- ***Photovoltaic route:*** solar cells -> DC electricity (rooftop, solar parks, pumps, lighting).
- Advantages: free, clean, abundant, no moving parts. Limits: ***intermittent*** (night, clouds), low density (needs large area), storage cost, high initial cost.

2. Solar Collectors – Working Principle

- A ***solar collector*** is a special heat exchanger that absorbs solar radiation and transfers the heat to a fluid (water, air, oil).

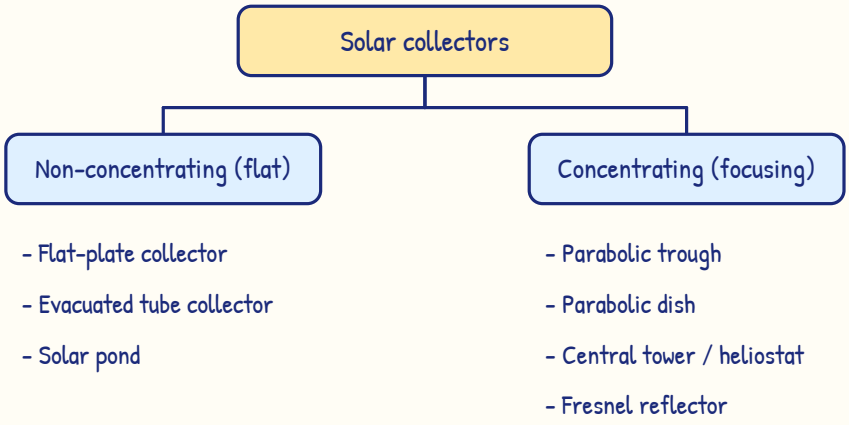


Fig: Classification of solar collectors

Flat-plate collector (FPC)

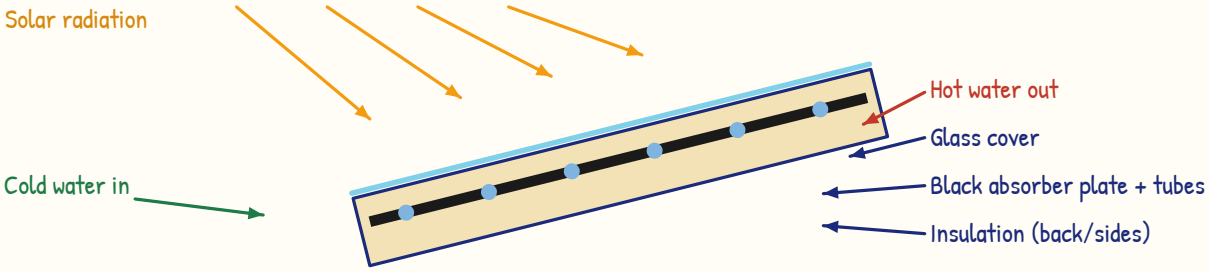


Fig: Flat-plate collector (cross-section)

- **Parts:** (1) **Black absorber plate** (copper / aluminium, selective coating -> high absorptivity, low emissivity); (2) **Tubes (risers + headers)** bonded to the plate carrying water; (3) **Transparent cover** (1 or 2 glass sheets) -- passes short-wave solar radiation but blocks the long-wave (**greenhouse effect**) and cuts convection loss; (4) **Insulation** (glass wool / PUF) at back and sides; (5) **Casing (box)**.
- **Working:** solar radiation passes the glass -> absorbed by plate -> heat conducted to tubes -> water heated (temperature 40-80 deg C). Natural (**thermosyphon**) circulation with tank above, or forced circulation with a pump.

$$\text{Useful heat } Q_u = A_c F_R [S - U_L (T_i - T_a)];$$
$$\text{Efficiency} = Q_u / (A_c I_T)$$

- S = absorbed radiation, U_L = overall heat-loss coefficient, F_R = heat removal factor, T_i = inlet fluid temperature, T_a = ambient temperature. **Tilt = latitude** facing south (in India).

Flat-plate	Concentrating
Uses beam + diffuse radiation	Only beam radiation; needs **sun tracking**
Temp. up to ~ 100 deg C	Temp. 150–1000 deg C
Low cost, simple, little maintenance	Costly; complex
Concentration ratio = 1	Concentration ratio 10–1000+

Concentrating collectors

- **Parabolic trough:** long curved mirror focuses sun on a receiver tube (oil, 400 deg C) -> steam -> turbine (SEGS plants). **Parabolic dish:** point focus, Stirling engine. **Central receiver (power tower):** field of **heliostats** reflects onto the tower boiler (up to 1000 deg C; molten salt storage).
- **Concentration ratio (C)** = aperture area / receiver area.
- **Other uses:** solar still (1 L/m^2/day), **solar pond** (salinity-gradient), solar cooker, **solar PV**: p-n junction, efficiency 15–22%; storage in batteries.

Example 1: Flat plate collector

- Collector area 2 m^2, incident radiation 800 W/m^2, efficiency 55%. Water flow 0.02 kg/s (c_p = 4.18 kJ/kg K).
- Useful heat = 0.55 x 800 x 2 = **880 W**. Temperature rise = 880 / (0.02 x 4180) = **10.5 deg C**.

3. Wind Energy

- Winds are caused by **uneven solar heating** of the earth (about 1–2% of solar energy converts to wind). Wind turbines convert the **kinetic energy of moving air** into rotation -> electrical energy.

$$\text{Power in wind } P = (1/2)\ \rho\ A\ V^3\ (W);\ \text{Power extracted} = (1/2)\ \rho\ A\ V^3\ C_p$$

- rho = air density (1.225 kg/m^3), A = swept area = pi D^2/4, V = wind speed. **Power depends on the cube of wind speed** -> site selection is critical.

- **Betz limit:** maximum possible power coefficient $C_{p,max} = 16/27 = 0.593$ (59.3%). Practical C_p = 0.35–0.45.
- **Tip speed ratio (TSR)** = blade tip speed / wind speed = omega R / V (4–8 for 3-blade); **cut-in speed** ~ 3–4 m/s; **rated speed** 12–15 m/s; **cut-out speed** ~ 25 m/s (turbine shuts down for safety).

Example 2

- Wind speed 12 m/s, rotor diameter 80 m, C_p = 0.4. Swept area = pi x 40^2 = 5027 m^2.
- P = 0.5 x 1.225 x 5027 x 12^3 x 0.4 = **2.13×10^6 W = 2.1 MW**.

Types of wind turbines

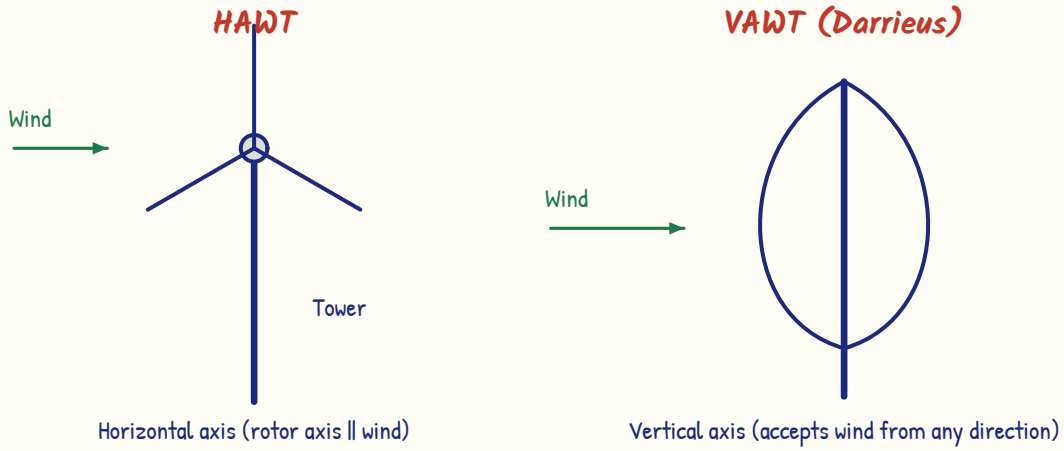


Fig: Horizontal-axis and vertical-axis wind turbines

Feature	HAWT	VAWT
Axis of rotation	Horizontal (parallel to wind)	Vertical (perpendicular to wind)
Yaw mechanism	Needed (turn into wind)	Not needed; accepts any direction
Efficiency	Higher (C_p 0.4–0.5)	Lower (0.3–0.4)
Types	1, 2, **3-blade** (standard), multi-blade (water pumping)	**Savonius** (drag type), **Darrieus** (lift type), H-rotor
Gearbox / generator	Placed at the top of tower (nacelle) -> maintenance difficulty	At ground level -> easy maintenance

Feature	HAWT	VAWT
Tower	Tall (80–120 m), needs strong wind	Short; suitable for urban / low wind
Self-starting	Yes	**Savonius yes; Darrieus no**
Usage	Large grid-connected wind farms	Small stand-alone, rooftop

Components of a HAWT

- **Rotor blades** (GRP / carbon fibre, aerofoil), **hub**, **pitch** system (blade angle control), **nacelle** containing low-speed shaft, **gearbox**, high-speed shaft, **brake**, **generator** (DFIG / PMSG / induction), **yaw drive**, **anemometer + wind vane**, **controller**, **tower**, foundation, transformer.
- **Control:** pitch control and stall control; yaw to face the wind; **variable-speed** operation with power electronics converters.
- **Wind farm:** array of turbines, spacing 5–10 D downwind and 3–5 D crosswind (wake effects); **offshore** wind has higher, steady wind.
- India: **~ 45 GW** (Tamil Nadu, Gujarat, Karnataka, Rajasthan, Maharashtra); Muppandal and Jaisalmer. Advantages: free, no emission, quick installation. Disadvantages: intermittent, noise, bird hazards, visual impact, site-specific, grid variability.

4. Tidal Energy

- Tides are the rise and fall of sea level caused by **gravitational attraction of the moon (and sun)** on the oceans. Two high and two low tides every day (period 12 h 25 min). **Spring tides** (new / full moon) -> large range; **neap tides** small range.
- Useful where the **tidal range > 5 m** and the shape of basin allows enclosing (e.g., Bay of Fundy, La Rance – France 240 MW, Sihwa – Korea; India: **Gulf of Khambhat, Gulf of Kutch, Sundarbans**).

> Energy per tide $E = (1/2)\ \rho\ g\ A\ R^2$ (J); Average power $P = E / T$ (T = tide period 12.4 h)

- A = basin area, R = tidal range, rho = 1025 kg/m^3 (sea water).

Example 3

- Basin area 10 km^2 (10^7 m^2), range 6 m. E = 0.5 x 1025 x 9.81 x 10^7 x 36 = **1.81×10^{12} J = 503 MWh** per filling. If only ~ 80% efficiency and generation is used over 6 h, average power ~ **70 MW**.

Types of tidal power plants

- ***Single-basin, single-effect:*** one basin; fills through sluices at high tide, ***generates at ebb*** only (4–5 h/day).
- ***Single-basin, double-effect:*** generates on both flood (inflow) and ebb (outflow) using ***reversible turbines***.
- ***Double-basin (two-pool):*** upper and lower basins give ***continuous / controlled*** output; costlier.
- ***Tidal stream / current turbines:*** like underwater wind turbines (no barrage, less environmental impact), tidal lagoons.
- Turbines: ***bulb***, ***rim (Straflo)*** and tubular types -- low head, axial flow; sluice gates; embankment; power house.
- Advantages: renewable, predictable, no fuel. Disadvantages: high cost, intermittent, low load factor (~ 25–30%), site-limited, disrupts marine life, corrosion of equipment, effect on silting.
- ***Ocean wave and OTEC*** (ocean thermal energy conversion using 20 deg C temperature difference between surface and deep water) are other ocean sources.

5. MHD Power Generation

- ***Magneto-Hydro-Dynamics:*** an electrically conducting ***fluid (ionised gas / plasma)*** moving across a ***magnetic field*** generates electricity directly (Faraday's law) -- ***no moving mechanical parts***.

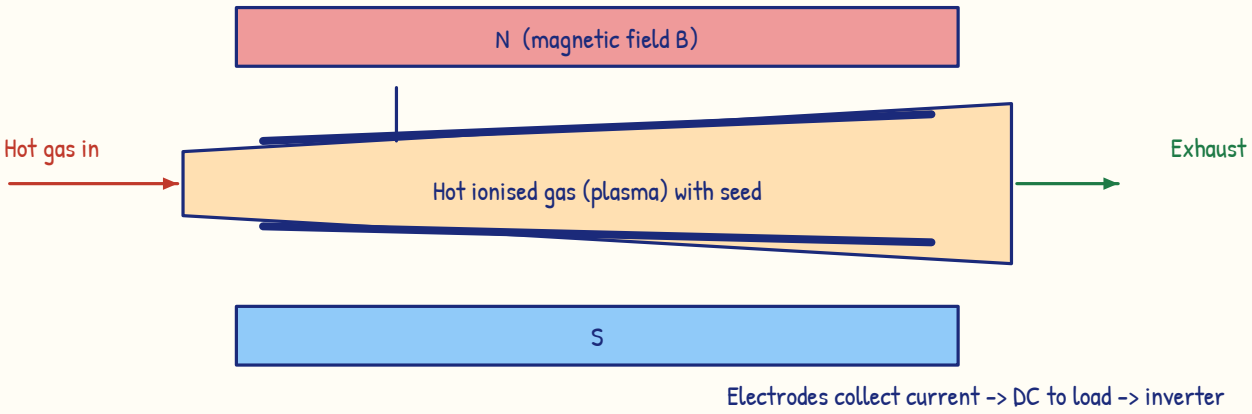


Fig: Principle of an open-cycle MHD generator

> *Induced emf $E = B\ v\ L$ (V), Power density = sigma $v^2\ B^2$ (W/m^3) (maximum at load factor 0.5)*

- Conductor: hot ***ionised gas*** from burning fuel at ~ 2500–3000 K, seeded with ***potassium or caesium*** (easy to ionise) to raise conductivity. The B field is produced by ***superconducting magnets***; electrodes collect DC current.

Types

- ***Open cycle:*** fuel (coal / oil / gas) burned with oxygen-enriched / preheated air, seeded, expanded through the nozzle into channel, then exhaust passes through ***air pre-heater, seed recovery***, and a steam bottoming cycle.
- ***Closed cycle:*** inert gas (helium / argon) + seed (caesium) circulates in a loop heated by a heat source (nuclear reactor, ~ 1800–2000 K). Liquid metal MHD (two-phase) also studied.

Advantages / disadvantages

- ***Efficiency 50–60%*** in ***combined MHD-steam*** plant (MHD alone 20%); quick start, no moving parts, less pollution, uses coal.
- Difficult materials (very high temp., electrode erosion), superconducting magnet cost, seed recovery, low conductivity; MHD is still at a ***pilot stage*** (e.g., in the U.S., Russia U-25, India).

6. Nuclear Power Station – Nuclear Fuel

- **Nuclear power plant:** heat released by **nuclear fission** in a reactor -> steam -> turbine -> generator (Rankine cycle). Only the heat source differs from a steam plant.
- **Mass defect and energy:** $E = \Delta m c^2$; 1 kg U-235 gives about 8.2×10^{13} J (= 2.5×10^6 kg coal); each fission releases ~ **200 MeV**.

Nuclear fuels

- **Fissile (directly fissionable by thermal neutrons):** U-235 (0.7% in natural uranium), **Pu-239, U-233**.
- **Fertile (converted into fissile by neutron capture): U-238 -> Pu-239, Th-232 -> U-233**.
- Natural uranium (0.72% U-235) used in **PHWR**; **enriched** uranium (2–5% U-235) in light-water reactors; fuel as **UO_2 pellets** in zirconium-alloy (zircaloy) clad tubes, bundled into fuel assemblies. Mining -> milling (yellowcake) -> conversion -> enrichment -> fabrication.

7. Nuclear Fission and Chain Reaction

- **Fission:** a heavy nucleus (U-235) absorbs a **slow (thermal, 0.025 eV) neutron** and splits into **two lighter fragments** (e.g., Ba-141 and Kr-92) + **2–3 neutrons** + energy (~ 200 MeV, mostly kinetic energy of fragments -> heat) and gamma rays.

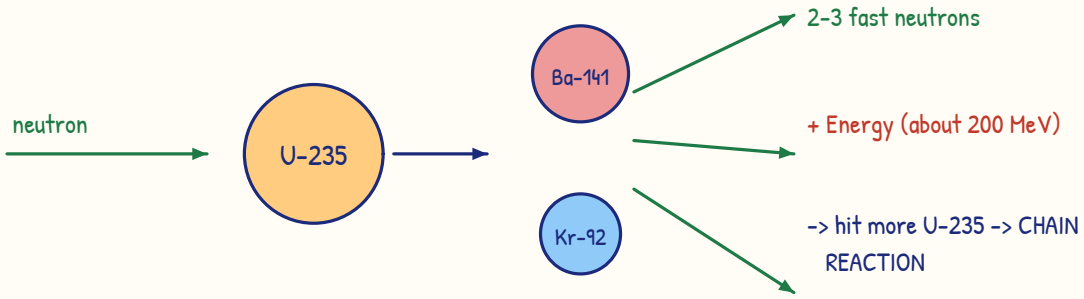


Fig: Fission of U-235 and release of neutrons

- U-235 + n -> Ba-141 + Kr-92 + 3 n + about 200 MeV (mass numbers: 235 + 1 = 141 + 92 + 3)

Chain reaction

- Neutrons released cause further fissions -> **self-sustaining chain reaction**. Measured by the **multiplication factor k = neutrons in one generation / neutrons in the previous generation**.

k value	State	Meaning
k < 1	**Sub-critical**	reaction dies out
k = 1	**Critical**	steady power (normal operation)
k > 1	**Super-critical**	power rises (start-up; bomb if uncontrolled)

- **Critical mass** = minimum fuel mass to sustain a chain reaction. Neutrons are lost by **leakage** and **non-fission capture**; the **four-factor formula** k_{inf} = eta epsilon p f; with leakage $k_{eff} = k_{inf} \times P_{NL}$.
- **Delayed neutrons (0.65%)** give time (seconds) to control the reactor mechanically.

Moderator

- Slows fast neutrons (2 MeV) to thermal (0.025 eV) energy by collisions: **light water, heavy water (D_2O), graphite, beryllium**. Good moderator = low atomic mass, high scattering and low absorption. Heavy water is best (allows natural uranium).

8. Breeding and Fertile Materials

- **Breeding** = producing more fissile material than is consumed. **Fertile materials** (U-238, Th-232) absorb neutrons and decay to fissile nuclei.
- **U-238 + n -> U-239 -> (beta decay) Np-239 -> (beta) Pu-239** (fissile). **Th-232 + n -> Th-233 -> Pa-233 -> U-233** (fissile).
- **Conversion ratio** = fissile atoms produced / fissile atoms consumed; if > 1 it is called **breeding ratio (BR)**.
- **Breeder reactor:** fast neutrons (no moderator), liquid sodium coolant, a core of Pu-239 surrounded by a **blanket of U-238**. **Fast breeder reactor (FBR)** BR ~ 1.2–1.4 -> converts the abundant U-238 into fuel ->

uses 60–70 x more energy from uranium. Doubling time 10–30 years.

- ***India's three-stage programme:*** Stage 1: ***PHWR*** (natural U) -> Pu. Stage 2: ***FBR*** (Pu + U-238; PFBR Kalpakkam 500 MW) -> U-233. Stage 3: ***Thorium-U-233 reactors*** (AHWR) -- uses India's huge thorium reserves.

9. Nuclear Reactor – Components and Types

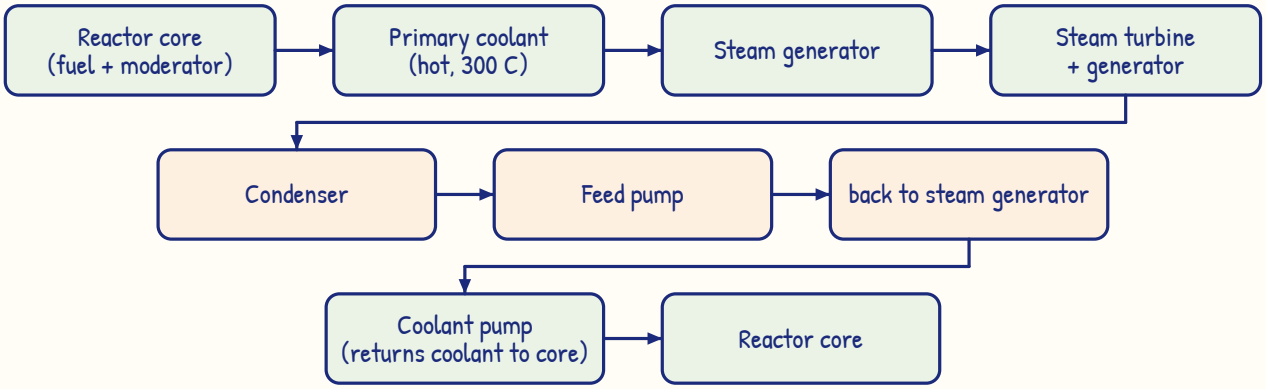


Fig: Block diagram of a nuclear power plant (PWR)

Main components of a reactor

- ***Fuel*** (UO_2 rods), ***moderator***, ***control rods*** (boron, cadmium, hafnium: absorb neutrons; ***shut-down / safety / regulating***), ***coolant*** (water, heavy water, CO_2, helium, liquid sodium), ***reflector*** (graphite / beryllium to reduce leakage), ***shielding*** (concrete + steel: biological shield), ***reactor pressure vessel***, ***containment building***.
- ***Coolant requirements:*** high heat capacity, low neutron absorption, non-corrosive, stable.

Reactor	Fuel	Moderator	Coolant	Features
PWR	Enriched UO_2	Light water	Light water at 150 bar (no boiling)	Most common; steam generator; secondary loop
BWR	Enriched UO_2	Light water	Light water boils in core (70 bar)	Direct steam to turbine; simpler; radioactive steam

Reactor	Fuel	Moderator	Coolant	Features
PHWR (CANDU)	Natural UO_2	Heavy water	Heavy water	On-load refuelling; used in India
GCR / AGR	Natural / enriched	Graphite	CO_2	High temperature
FBR	Pu + U	None (fast neutrons)	Liquid sodium	Breeds fuel
HTGR	Coated particle	Graphite	Helium	800–950 deg C; hydrogen production

- Indian stations: **Tarapur** (BWR 1969), **Rawatbhata**, **Kalpakkam** (PHWR, PFBR), **Narora**, **Kakrapar**, **Kaiga**, **Kudankulam** (VVER-1000 PWR, Russian), Gorakhpur. Capacity ~ 8 GW.

10. Reactor Operation

- **Start-up:** control rods withdrawn slowly -> neutron flux rises (monitored by **neutron detectors** -> source range -> intermediate -> power range) until k = 1 (critical) -> power raised to the desired level and turbine coupled.
- **Power control:** inserting control rods reduces k (and power); withdrawing raises power. Also **boric acid** (soluble poison in PWR) for slow control; coolant flow and temperature feedback.
- **Shutdown:** all rods (**SCRAM**) inserted automatically on abnormal conditions (high power, high temperature, low flow) within seconds. After shut-down, the **decay heat** (~ 7% initially) still needs cooling -> **emergency core cooling system (ECCS)**, passive cooling.
- **Self regulating features: negative temperature coefficient** and void coefficient make the reactor stable.
- **Poisons:** Xenon-135 and Samarium-149 absorb neutrons -> affect restart soon after shut-down (**xenon pit**).
- **Fuel burn-up** ~ 7000 MWd/t (PHWR) to 40,000–60,000 MWd/t (PWR). Refuelling: on power (CANDU) or shut-down (PWR, every 12–18 months). Spent fuel -> **cooling pond** -> reprocessing (**PUREX**) -> waste

vitrification and geological disposal.

Safety and radiation

- **Defence in depth:** fuel clad -> pressure boundary -> **containment** (reinforced concrete). Dose limits: public 1 mSv/year; radiation workers 20 mSv/year (averaged). **AERB** regulates in India.
- Accidents: Three Mile Island (1979), **Chernobyl (1986)**, **Fukushima (2011)** -> improved passive safety.

Advantages of nuclear power	Disadvantages
Tiny fuel quantity; very high energy density	High capital cost; long construction
No CO_2 / SO_2 / ash; reliable **base load** (CF > 80%)	Radioactive waste (long-lived); decommissioning
Large reserves with breeders / thorium	Accident risk, public acceptance, security
Low fuel cost	Needs skilled staff, cooling water; thermal pollution; poor load following

Example 4: Fuel consumption

- A 300 MW (electrical) plant, efficiency 30%, so thermal power = 1000 MW. Energy per fission 200 MeV = 3.2×10^{-11} J.
- Fissions per second = $10^9 / 3.2 \times 10^{-11}$ = **3.1×10^{19} /s**. Mass of U-235 consumed per day = $3.1 \times 10^{19} \times 86400 \times 235 \times 1.66 \times 10^{-27}$ kg = **~ 1.05 kg / day**.
- Compare coal: 1000 MW(t) needs about **3000+ tonnes of coal per day**.

Q & A (exam revision)

- **What is the solar constant?** 1367 W/m^2 on a surface normal to the sun's rays outside the earth's atmosphere.
- **Why is glass cover used in a flat-plate collector?** It transmits short-wave solar radiation but traps the long-wave heat from the plate (greenhouse effect) and reduces convective loss.

- **Why does a wind turbine have a cut-out speed?** To protect the rotor, gearbox and tower from damage in storms (about 25 m/s).
- **Why seeding in MHD?** Gas at 2500–3000 K alone has low conductivity; alkali metal (K, Cs) ionises easily and raises conductivity.
- **Role of moderator vs control rod?** Moderator slows neutrons to thermal energy; control rods absorb neutrons to regulate k.
- **Why heavy water for natural uranium?** Heavy water absorbs very few neutrons and moderates well, so enough neutrons remain to sustain the chain reaction without enrichment.
- **What is a SCRAM?** Rapid automatic insertion of all control rods to shut down the reactor.

Quick revision

- Solar constant 1367 W/m^2. FPC: black plate + glass cover + insulation. Wind power = 0.5 rho A V^3; Betz limit 59.3%.
- Tidal energy = 0.5 rho g A R^2; need range > 5 m. MHD: conducting gas + magnetic field -> direct DC; 50–60% in combined cycle.
- Fission: U-235 + n -> fragments + 2–3 n + 200 MeV. k = 1 critical. Moderator slows, control rods absorb, coolant carries heat.
- Fertile: U-238, Th-232; fissile: U-235, Pu-239, U-233. FBR breeds fuel (BR > 1).